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A guide to

Women's Health: *Fifty and Forward*



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A GUIDE TO WOMEN'S HEALTH: FIFTY AND FORWARD

SPECIAL HEALTH REPORT

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Dear Reader,

When you were younger, most of the health decisions you made probably seemed pretty straightforward—for example, what to eat, when to exercise, and how long to sleep. If you got sick, you worked with your doctor to treat the illness. At midlife, however, medical decisions become more complicated, and the stakes are higher. Should you use hormone therapy? Are you doing enough to keep your bones strong? Should you worry about those memory slips that seem to be cropping up more regularly?

This report is here to help. In the following pages, you will find information about common conditions that affect most women at this time of life, from menopausal symptoms to flagging energy—and, more importantly, how to cope with these issues. This report also covers the major diseases that can afflict women, and it gives you evidence-based information on how to help prevent them. (For simplicity, we use the term “women” throughout to refer to individuals assigned female at birth; this language does not imply any assumptions about gender identity.)

The good news is that many things are within your control. For example, following a plant-based eating plan may reduce your risks for a number of problems, including heart attack, stroke, diabetes, and Alzheimer’s disease. Women who exercise regularly are less likely to feel depressed and anxious and less likely to develop a host of chronic debilitating conditions. And did you know that getting sufficient sleep—from seven to nine hours a night—will not only boost your energy, but also help you lose weight and keep it off? Decisions that you make about your health now may directly affect how you feel, physically and emotionally, for decades to come.

There is no single formula for health that works for everyone. Each of us is a bit different. A test that was recommended for a friend may not be necessary for you, depending on your family history. A medication that works wonders for a colleague might not do the trick for you. So it’s important, in addition to reading and learning about health, to develop trusting relationships with health care providers who really get to know you, who can individualize recommendations for you, and who can make you feel comfortable about discussing sensitive information.

This report is designed to guide you to a healthier place by providing information that will help you work together with your clinician to make great choices. The information contained here is intended to help you sift through what’s important and empower you to take whatever steps make sense for you to lead the happiest, healthiest, and safest life for many years to come.

Sincerely,

Toni Golen, M.D.

Medical Editor

Planning for a healthy future

Midlife can be a woman's halftime celebration. Not only is it an opportunity to reflect on and rejoice in the life you've lived so far, but it's also a good time to plan your strategy for the future.

If you're like most women, menopause is likely to give you a new perspective on life and health. Premenstrual syndrome, menstrual discomforts, and reproductive issues are no longer a concern. Hot flashes and other perimenopausal symptoms are gradually disappearing. As you move into the sixth decade of life, you may be feeling the burst of energy and optimism that anthropologist Margaret Mead famously termed "postmenopausal zest."

In the decades to come, your health focus will continue to shift. Initially, you are likely to be most concerned with preventing or managing the conditions that become more prevalent as estrogen levels decline—for example, osteoporosis, heart disease, sexual dysfunction, and incontinence. Once you have reached your eighth or ninth decade, you may be more focused on staying strong enough to live independently, pursue the hobbies that give you pleasure, and enjoy your family and friends.

Every step of the way, the choices you make can have a real impact on your health. So how do you start planning for a healthy future? First, acknowledge what you can't control, such as your genetic risks. Then put your energy into changing what you can, such as making healthier lifestyle and diet choices.

What you can't control

Even though you can't change the following factors, knowing your risks can help you plan for the future and stay as healthy as possible.

Age. As people grow older, their bodies undergo gradual physical changes that are normal and inevi-



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Midlife is a time to rejoice in your life and plan for a healthy future. Aging is not a disease, and illness is not the inevitable result of aging.

table. Such changes begin with random genetic mistakes that occur as cells divide and DNA is improperly copied. Over time, the cellular damage accumulates, affecting every tissue and organ of the body. These changes occur at different rates throughout the body and from person to person.

Family medical history. If an immediate family member—for example, a parent or sibling—developed a problem such as heart disease or cancer, especially if it occurred early in life, it could mean that you are at risk as well. It's important to know what steps you can take to decrease your risk of certain conditions, even if a number of people in your family have been affected in the past.

Race, socioeconomic status, and other social determinants of health. Social determinants of health are non-medical factors that can have a major impact on your health and well-being, yet most are beyond personal control. For instance, we now know that systemic racism contributes to an increased risk of heart disease in black women. Discrimination may similarly lead to an increased risk of cardiovascular disease in gay and bisexual women. Other social determinants of health include access to health care and nutritious foods and a safe and stable living environment.

What you can control

It may surprise you to know that what you can control often affects your health much more than the factors that are beyond your control. Genes are powerful, but many of them are responsive to lifestyle changes such as improvements in diet and exercise. As you make these changes, disease-promoting genes become less active, while genes that protect you become *more* active. Here are just a few lifestyle factors to consider.

Whether you smoke. Roughly 10% of American women still smoke. If you are one of them, kicking the habit may be the single most important thing you can do to improve your health, simultaneously reducing your risks of heart disease, stroke, and various cancers. Some women have switched to e-cigarettes (vaping devices). However, the FDA has not approved these devices as a smoking cessation aid, and the CDC maintains that no tobacco products are safe. Vaping products also present their own dangers, including exposure to contaminants that can lead to serious lung damage.

What you eat. Regularly consuming a healthy diet is one of the best ways to help prevent—and manage—heart disease, high blood pressure, and diabetes. Doing so may even reduce your risk of dementia.

How much you move. Get active, live longer—and better. Study after study has linked greater amounts of physical activity to improved health. And this doesn't just include time spent at the gym working out. Even daily activities, such as carrying in the groceries or weeding the garden, bring health benefits.

In this report

This report will take a look at all these factors and more, how they affect your health, and what you can do to prevent or manage health problems.

First, we'll look at some of the most common health conditions that women start to develop in middle age, along with medical issues specific to women, and what you can do about them. The health concerns that most people start to encounter in midlife fall into five broad categories, which we address one by one in this report.

Hormonal, sexual, and urinary issues. As you enter menopause, changes in hormone levels in your body may lead to symptoms such as vaginal dryness,

which may interfere with sexual function. Your sexual organs may shift position, and you may also experience urinary or fecal incontinence. While these symptoms aren't usually dangerous to your health, they can greatly affect your quality of life.

Vision and hearing problems. In a condition called presbyopia, the eye's natural lens stiffens somewhat, making it harder to read closely held materials like books and menus. Tear production declines, leading to dry eye syndrome. Cloudiness in the lens (cataract) can blur vision. Hearing impairment can make it more difficult to follow conversations in noisy environments.

Musculoskeletal conditions. Bones may become weaker and more prone to fracture, and the padding in your joints may thin, causing the bones to rub together and causing a painful condition called osteoarthritis. Your feet can develop a variety of ailments after decades of standing, walking, and running—not to mention wearing poorly designed shoes. If not managed, these musculoskeletal problems can make it harder for you to move around.

Changes in the mind and memory. As we age, it's common to experience more forgetfulness, although this isn't necessarily an early sign of dementia. There are practical ways to adapt to age-related memory slips. Depression and anxiety disorders can also emerge.

Less energy and muscle power. Many women report low energy and fatigue in middle age and beyond. For some, it's a warning sign of thyroid trouble, but for many more, it isn't. Getting more exercise and better sleep can boost energy levels, while muscle-building exercise in particular can reverse or slow a loss of strength.

After reviewing these common ailments, we'll take a deeper dive into the factors that determine your risk for major illnesses—such as heart disease, diabetes, and breast cancer—that can cut life short and compromise your active, independent lifestyle.

Finally, our Special Section (page 42) lays out 10 steps you can take to reduce your risks for these health problems. Remember, aging is not a disease, and illness is not the inevitable result of aging. Use this report as a guide to evaluating your health portfolio and making investments—not only to preserve your current health, but also to feel your best. ♥

Meeting midlife challenges

There are many serious diseases that can hamper your life as you get older. Those conditions may or may not affect any given individual. But a host of lower-level problems—things like vision changes, joint pain, and reduced energy levels—affect most women in their 50s and beyond. Some of them stem from declining levels of estrogen, which helps to maintain tissues in

the body's reproductive organs as well as in the breasts, brain, bones, bladder, and cardiovascular system. These conditions might not rob your life of years, but they can rob your years of life. Some of these ailments are preventable, and some are curable, but all are treatable.

These conditions fall into five broad categories. We'll address each one in turn.

Hormonal, sexual, and urinary issues

Persistent menopausal symptoms

Although some women breeze through menopause, others have symptoms that persist for years. A report from the Study of Women's Health Across the Nation, or SWAN, a long-term study of women of different races and ethnicities who are in the menopausal transition, found that up to 80% of women experience symptoms such as hot flashes, night sweats, and heart palpitations, and that those symptoms in some cases last more than eight years. Other common symptoms are sleep disruptions, an increase in depression and anxiety, and vaginal dryness. Hormone therapy (HT) is perhaps the best-known treatment, but there are other strategies that can help.

Hormones or not?

If you have severe hot flashes, nothing delivers more effective relief than HT, which is typically a combination of estrogen and progestin. (You must use both unless you've had a hysterectomy, since taking estrogen alone raises your risk of endometrial cancer—that is, cancer of the uterine lining.) But it's important to weigh the pros and cons. Your decision largely depends on three things: the severity of your symptoms, whether nonhormonal treatments have been effective for you, and whether you have any contraindications, meaning reasons why this particular treatment might be especially risky for you.

More than 20 years ago, the Women's Health Initiative (WHI)—the largest clinical trial of postmenopausal hormone use—found that HT increased the risk of heart attacks, strokes, deep-vein thrombosis, pulmonary embolisms, and breast cancer. However, more recent research suggests that it is not as risky as the WHI researchers initially indicated, in part because the study included many women *without* menopausal symptoms as well as a significant num-

Table 1: Low-dose vaginal estrogen therapy for vaginal atrophy

These products, which contain low doses of estrogen, can reverse tissue thinning, dryness, and other age-related changes of the vagina without significantly increasing blood estrogen levels. Although the package inserts for these products currently describe the same risks as for systemic hormonal medications, the risks are in fact much lower, since only minimal amounts of hormone are absorbed into the bloodstream.

FORM	TRADE NAME(S)	HOW IT'S USED
Cream	Estradiol cream, Estrace, Premarin	Small amount of cream inserted into the vagina on a prescribed schedule
Ring	Estring	Small ring inserted into the vagina once every three months
Tablet	Vagifem, Imvexxy, Yuvaferm	Small tablet inserted into the vagina nightly for two weeks at the start of treatment and then twice a week

What about bioidenticals?

Bioidentical hormones—which the Endocrine Society defines as “compounds that have exactly the same chemical and molecular structure as hormones that are produced in the human body”—are prepared at compounding pharmacies according to a clinician’s prescription and touted as “natural” alternatives to “synthetic” brand-name drugs. Advocates imply that the hormones provided by compounding pharmacies are more precise copies of a woman’s hormones than those produced by pharmaceutical companies, but this is incorrect. In fact, many of the FDA-approved estrogen products on the market today fit the definition of bioidentical.

What’s more, they’re made from the same ingredients and by the identical suppliers that the compounding pharmacies use.



Furthermore, despite advertising claims, there is very little medical evidence that bioidenticals are any more beneficial than Premarin or Prempro. And, unlike the FDA-approved bioidenticals, those from compounding pharmacies haven’t undergone the scrutiny and clinical testing required for approval by the FDA, nor do they come with the same guarantees of dose and purity.

ber of women in their 70s. (Many experts believed that HT could prevent chronic diseases, which largely did not pan out.)

More recent studies and analyses show that increases in cardiovascular problems, which were slight, mainly impacted women who began taking hormones more than 10 years after menopause. Today, the North American Menopause Society says that “the benefits of hormone therapy outweigh the risks for most healthy women [with bothersome symptoms] who are aged younger than 60 years and within 10 years of menopause onset.”

Still, you should know that a combination of estrogen and progesterone raises the risk of breast cancer in menopausal women of all ages, especially for those who take it for many years. That’s why women who have a history of breast, uterine, or endometrial cancer—all of which can be driven by estrogen—are usually advised to avoid HT.

Estrogen also increases the risk of seizures, so if you have epilepsy, HT is not a good bet for you. And anyone with a history of liver problems or blood clots should not take estrogen in pill form. (The estrogen in oral formulations is processed in the liver, which, in response, increases its production of natural substances that promote clotting).

If you don’t have any contraindications and you have moderate to severe symptoms—such as debilitating hot flashes and night sweats that rob you of sleep—systemic HT may be a wise choice. Still, doctors advise using the smallest effective dose for the shortest possible time needed to obtain relief.

Hormones are available in many forms, including pills, patches, creams, and sprays (see Table 2, page 6), and most work systemically (throughout the body). If you primarily have vaginal symptoms (such as dryness, burning, and pain during sex), then low-dose vaginal estrogen may be sufficient (see Table 1, page 4). It’s applied locally, and very little gets into the rest of the body. Breast cancer survivors who used it for 5 years had no increased risk of recurrence, according to a 2023 study published in *Obstetrics & Gynecology*.

Nonhormonal approaches

Whether or not you’re a candidate for HT, there are other options that may be worth considering.

Non-hormonal medications. In 2023, the FDA approved fezolinetant (Veoza), the first nonhormonal drug designed to reduce hot flashes. If you take it, your doctor should monitor your liver function with blood tests, because fezolinetant can cause rare but serious liver injury (one case reported to the FDA). A similar drug called elinzanetant has shown promising results in late-stage clinical trials.

Other nonhormonal prescriptions drugs used to treat hot flashes include certain antidepressants, blood pressure medications, and drugs that treat nerve pain. Antidepressants may be used solely to treat hot flashes, or they may do double duty and treat depression as well. Brisdelle (paroxetine) is the only antidepressant that’s FDA-approved to treat hot flashes; it contains the same drug found in Paxil, but at a lower dose. Other antidepressants, including citalopram can also be effective for hot flashes.

Table 2: Hormone therapy for menopausal symptoms

These products increase blood levels of estrogen and treat both vaginal dryness and other menopausal symptoms such as hot flashes and night sweats. Women without a uterus can take estrogen by itself; those who have a uterus must take progestin in addition to estrogen (either separately or in a combination product) to minimize the risk of uterine cancer that can occur with estrogen alone. (The term “progestin” refers to the family of hormones that includes progesterone. Medications may include different types of progestins, all of which act in a similar way on the body and counteract the effect of estrogen on the growth of the uterine lining.)

HORMONE(S)	TRADE NAME
Oral estrogen	
17-beta estradiol	Estrace
conjugated equine estrogens	Premarin
esterified estrogens	Menest
conjugated estrogens/ bazedoxifene	Duavee
Transdermal estrogen	
17-beta estradiol	Alora, Climara, Menostar, Minivelle, Vivelle-Dot (patch); Divigel, Elestrin, Estrogel (gel); Evamist (spray)
Vaginal estrogen	
estradiol acetate	Femring (ring)
Oral progestin	
medroxyprogesterone acetate	Provera
norethindrone	Generic only
Intrauterine progestin	
levonorgestrel	Mirena (intrauterine device)
Oral progesterone	
micronized progesterone	Prometrium
Oral combination estrogen-progestin	
17-beta estradiol plus norethindrone acetate	Activella
bioidentical estrogen plus bio- identical progesterone	Bijuva
conjugated estrogens plus medroxyprogesterone acetate	Premphase 14/14, Prempro
estradiol plus drospirenone	Angeliq
ethinyl estradiol plus norethindrone acetate	Femhrt
Transdermal combination estrogen-progestin	
17-beta estradiol plus levonorgestrel	Climara Pro (patch)
17-beta estradiol plus norethindrone acetate	CombiPatch (patch)

Source: U.S. Food and Drug Administration.

Lubricants and moisturizers. They won't help with hot flashes, but they can relieve dryness and itching and make sex more comfortable. Lubricants like Astroglide and K-Y Sensual Silk or Ultra Gel are water-based and pH neutral. They approximate the feel of vaginal secretions and provide temporary relief. Moisturizers like K-Y Liquibeads, Luvena, and Replens cling to the cells lining the vaginal wall for longer-lasting effectiveness.

Moisture-wicking clothing. Moisture-wicking clothing and pajamas can be helpful for women prone to hot flashes as well as night sweats (hot flashes that occur while you're sleeping). Dressing in layers you can shed easily may also be helpful.

Exercise. Women who are active tend to suffer fewer hot flashes. Exercise can also give you a mood and memory boost and help you sleep more soundly.

Temperature control. Turn down the thermostat at night to reduce the risk and severity of night sweats. You may also want to keep one room in your home cool during the day so you can retreat to it when you feel a hot flash start, or keep a small fan handy. Plunging your hands in cold water and then patting your face, neck, and chest may also help.

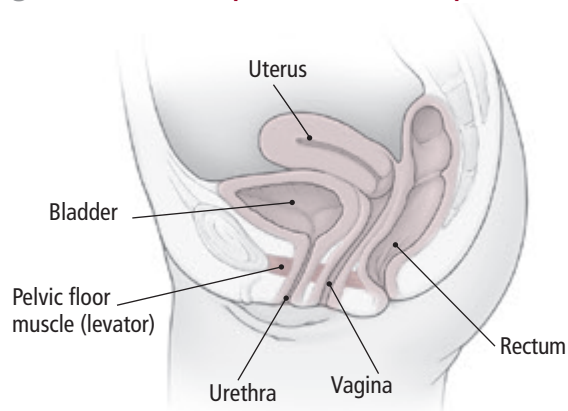
Stress management. Meditation, mindfulness, and deep breathing practices may help mitigate hot flashes as well as the cognitive fuzziness (“brain fog”) that many menopausal women experience.

Pelvic organ prolapse

As many as one in every five women has some kind of pelvic organ prolapse—a condition in which uterine, bladder, urethral, or rectal tissue protrudes into the vagina (see Figure 1, page 7). This occurs because the connective tissue (called ligaments and fascia), as well as the muscles of the pelvic floor, weaken over time. Symptoms may include a bulge or vaginal pressure, along with urinary, defecatory, or sexual dysfunction. It is one of the most common causes of urinary incontinence.

The risk of pelvic organ prolapse increases with age, and it's more common in women who've given birth, either vaginally or by cesarean (though this problem is more common among those who've given

Figure 1: Normal positions of the pelvic organs



In pelvic organ prolapse, the pelvic floor muscles and connective tissue that support the organs are weakened. One or more of the organs shift from their normal positions and can press against the walls of the vagina or protrude through it.

birth vaginally). Women who have had more than one vaginal childbirth are at highest risk. Giving birth to a baby that weighed more than 8.5 pounds is also thought to increase risk.

Another contributing factor is hormonal changes during menopause—specifically, a decline in estrogen, although experts aren't entirely sure why this plays a role. Other risk factors include a family history of the condition, obesity, chronic constipation (which causes frequent straining during bowel movements), and age. Nearly 40% of women with pelvic organ prolapse are 60 to 79 years old, and half are 80 or older.

Treating pelvic organ prolapse

Pelvic organ prolapse may produce no symptoms, in which case it does not need any treatment at all. For women who have symptoms such as urinary incontinence, one option is to use a pessary, a device similar to a diaphragm that's inserted in the vagina to help support the pelvic area. Other techniques that may help are Kegel exercises, which involve contracting and relaxing pelvic floor muscles, and physical therapy involving biofeedback or a massage-like technique called myofascial release.

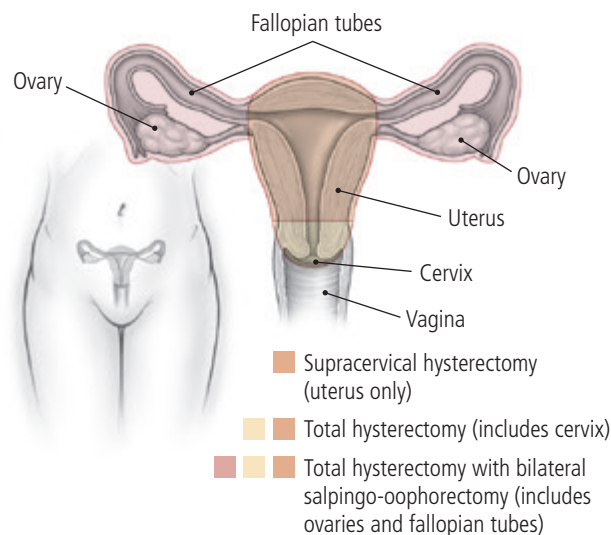
Surgery to strengthen connective tissues supporting pelvic organs and areas of weakened vaginal tissue is an option for women who suffer from symptoms of pelvic organ prolapse.

Hysterectomy. For women who have completed childbearing, hysterectomy may be an option. This procedure is often combined with ligament and tissue repair as well as procedures to address any associated incontinence (see “Treating urinary incontinence,” page 8), since removing the uterus alone may not eliminate symptoms associated with pelvic organ prolapse.

Hysterectomies used to require major open surgery, but today that's rarely the case. Most hysterectomies are now done laparoscopically using tiny cameras and surgical instruments inserted through small incisions in the abdomen, sometimes with the assistance of a robot, or through the vagina (no abdominal incisions needed). Such minimally invasive techniques allow for quicker recovery. The approach depends upon the reason for the hysterectomy, your anatomy, and your medical issues, as well as a balancing of risks and personal preferences.

Should you also remove the ovaries and fallopian tubes? The ovaries were once routinely removed when postmenopausal women were having a hysterectomy (see Figure 2), and they still are when a patient has a

Figure 2: Types of hysterectomy



In older women, most hysterectomies are usually performed for one of two reasons: pelvic floor prolapse or cancer (including precancerous conditions). The ovaries may be removed along with the uterus in order to prevent or treat ovarian cancer, but this will interrupt the production of small amounts of testosterone, which continues after menopause.



If you're running to the bathroom often or having trouble getting there in time, talk to your doctor about treatment options.

gynecologic cancer. The decision to remove the ovaries and fallopian tubes when hysterectomy is done for other reasons, such as prolapse, has potential benefits as well as risks. After menopause, the ovaries continue to produce small amounts of hormones such as testosterone, which may play a part in sex drive, into the eighth decade of life. However, removing them reduces the risk of ovarian cancer. This is a complex decision that is best made after discussion with your doctor.

Incontinence

The unintended loss of urine or feces can make it difficult to maintain good hygiene and carry on ordinary life. Besides disrupting daily activities and nighttime sleep, incontinence can also chip away at your health. If you have stopped exercising for fear of leakage, for example, you are giving up one of the most effective ways of maintaining health. If you're getting up several times each night to go to the bathroom, you're risking not only sleep deprivation but also injury from falls. Older women who frequently rush to the bathroom are 26% more likely to fall and 34% more likely to break a bone. The good news is that there are a number of effective medications, as well as procedures that can be done with minimally invasive techniques.

Urinary incontinence typically takes the form of stress incontinence or urge incontinence, though some women have both. Stress incontinence refers to urine leaking during exercise, sneezing, or coughing. It is often due to weakening of the pelvic floor muscles or the urinary sphincter (the muscle that

controls the opening of the bladder). Urge incontinence occurs when the bladder contracts suddenly, often before it is full (a condition also known as overactive bladder). People with urge incontinence frequently experience a sudden and overwhelming urge to void and may leak urine if they can't get to a bathroom in time.

Treating urinary incontinence

Several approaches have been successful in relieving urge incontinence, stress incontinence, or both:

Bladder control training. Women with urge incontinence practice consciously suppressing the urge to urinate for progressively longer intervals.

Fluid management. This approach is based on minimizing alcohol, caffeine, and carbonated beverages; restricting fluids to less than 64 ounces daily; and drinking slowly and no more than 8 ounces at a time.

Pelvic muscle strengthening. Exercising the pelvic muscles by contracting them and releasing them can be performed actively by doing Kegel exercises (tighten the muscles you would use to avoid passing gas), or passively, with either electrical stimulation or pulsed magnetic fields. Biofeedback training may help you to locate and contract these muscles.

Weight loss. Studies have shown that weight loss helps alleviate symptoms, although researchers are not sure why. It may reduce pressure on the bladder.

Medication. Vaginal estrogen can help relieve urinary incontinence if vaginal atrophy is present. There are currently no FDA-approved medications for stress incontinence, but the vaginal estrogen ring (Estring; see Table 1, page 4) is approved for some urinary symptoms. A class of drugs called antimuscarinics, which block nerve impulses to the bladder, can help with urge incontinence. However, research has highlighted an association between these drugs and cognitive decline, especially in older adults who use them for an extended period. Your doctor can help you weigh the pros and cons. Another type of medication, mirabegron (Myrbetriq), also treats urge incontinence. It relaxes the bladder and allows it to fill more completely, and it has not been linked to cognitive problems.

Pessaries. A pessary is a device made of silicone that comes in a variety of shapes and sizes and is fitted

into the vagina, much like a diaphragm. Left in place there, the pessary compresses the urethra against the upper rear portion of the pubic bone, which can help prevent urinary leakage. However, pessaries designed to relieve symptoms of pelvic organ prolapse (see “Treating pelvic organ prolapse,” page 7) sometimes worsen urinary incontinence.

Botulinum toxin A. Injections of botulinum toxin A (Botox) are FDA-approved for treating overactive bladder when oral medications to inhibit bladder contractions can’t be taken or are ineffective.

Electrical stimulation. Two treatment options for urge incontinence use nerve stimulation to control bladder contractions. One approach, called percutaneous tibial nerve stimulation, involves using a device to send mild electrical pulses to the tibial nerve, just under the skin near the ankle. These pulses travel to the sacral nerve plexus, the group of nerves at the base of the spine responsible for bladder function. Because improvement happens gradually, the procedure is performed in a series of half-hour office visits. A second approach stimulates the sacral nerve through a permanently implanted device that resembles a cardiac pacemaker, placed under the skin in the buttock area. Both approaches are moderately successful for alleviating urge incontinence.

Injections. Collagen and other bulking agents can be injected around the urethra to tighten the seal between it and the bladder. The procedure is used primarily for stress incontinence. Urethral injections often need to be repeated and are usually reserved for those who can’t or won’t undergo surgery.

Surgery. Bladder-suspension procedures raise the bladder and urethra with a sling of tissue or synthetic mesh to prevent urine from leaking. At least two sling procedures can be performed through tiny incisions under local anesthesia. (In 2019, the FDA banned the use of transvaginal mesh products for treating pelvic

organ prolapse. However, the use of synthetic mesh sling surgery for stress urinary incontinence is still considered safe and effective.) Hysterectomy may be done at the same time if incontinence is accompanied by uterine prolapse (see “Hysterectomy,” page 7).

Treating fecal incontinence

Treatments for fecal incontinence aim to control diarrhea and constipation by adding more fiber to the diet, strengthening the pelvic muscles, and improving the ability of the anal sphincter to contract.

Pelvic muscle strengthening. Kegel exercises can strengthen pelvic floor and sphincter muscles. Bio-feedback training offers much the same benefit, but, in addition to helping you strengthen and coordinate the action of your sphincter muscles, it can improve your ability to sense the presence of stool in the rectum. It may offer up to a 75% chance of improvement and a 50% chance of complete relief.

Electrical stimulation of pelvic floor muscles (sacral neuromodulation). This technique uses a surgically implanted device to stimulate the sacral nerves, which are located in the lower back and serve the bladder, bowel, and pelvic floor. This device increases bladder and bowel sensations to improve control. One study published in the *Journal of Neurogastroenterology and Motility* found that the implant reduced fecal incontinence in 88% of people who received it.

Surgery. If you require surgery, the exact approach will depend on the cause of your symptoms and your response to previous treatments. The operations include procedures to correct rectal prolapse; to repair and raise the pelvic floor muscles; and to reshape, repair, and replace the anal sphincter. Colostomy is a very last resort. This procedure involves bringing the end of the intestine to a surgical opening in the patient’s abdomen. Feces are collected in a small pouch worn over the opening.

Vision and hearing problems

Presbyopia

As we age, the eye’s natural lens starts to stiffen and lose its ability to change shape in order to focus on nearby objects. This makes it more difficult to read and per-

form detailed tasks that require hand-eye coordination, such as threading a needle. The medical term for this condition is presbyopia, and it’s the most common age-related vision impairment in people 40 and older.

Treating presbyopia

The easiest way to address presbyopia is with glasses or contact lenses. If your distance vision is good, all you need is a pair of readers (over-the-counter reading glasses). If you're nearsighted, then prescription bifocals, trifocals, or progressive lenses can give you a mix of near, mid-range, and distance vision correction.

Contact lens wearers can try monovision correction. This means wearing a contact lens for distance vision in one eye (for use in driving and walking around) and a different contact lens for near vision in the other. Some people do well with monovision correction, but others have difficulty adapting to it.

More invasive options include:

- Multifocal lens implants can focus your vision at a range of distances simultaneously. That means if you receive an artificial lens as part of a cataract operation (see "Treating cataracts," page 11), you can get your presbyopia corrected, too.
- LASIK and PRK surgery use a tiny laser beam to sculpt the surface of the cornea, the transparent protective dome in the front of the eye. This can correct near vision for reading, although you'll probably need eyeglasses for distance vision. It's also possible to correct one eye for near vision and the other for distance vision (monovision correction).

Dry eye syndrome

As you age, your tear production declines. The resulting dryness can cause irritation, burning, or a slightly painful, scratchy feeling in the eye. Sometimes mucus accumulates, producing a sticky sensation. You may become sensitive to light, have trouble wearing contact lenses, or even find it difficult to cry. Dry eye can also cause fluctuations in vision or blurred vision. Studies have found that women are 70% more likely to have dry eye syndrome than men.

Treating dry eye syndrome

If your dry eye is caused by a reduction in natural tear production, eye drops may help. For moderate to severe cases, treatment usually involves some type of topical medication. There have been several eye drop recalls in recent years, so it's a good idea to



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A comprehensive eye exam is essential for diagnosing vision problems, including cataracts and other issues that become more common with age.

consult your eye doctor, a pharmacist, or the FDA website (www.fda.gov) before choosing a product. When dry eye is very severe, doctors may prescribe tears made of your own serum, the clear liquid part of your blood.

There are also drug-free ways to treat dry eye. One option is to use TrueTear, an FDA-approved handheld device that emits small pulses of energy to temporarily stimulate tear production. TrueTear has been shown to work well for producing tears, but because you place the device in your nose, it can cause side effects like nasal pain, nosebleeds, and sneezing.

Another way to combat dry eye is to break the habit of staring at your computer screen for hours at a time. Follow the 20-20-20-20 rule: Every 20 minutes, take a 20-second break and look away from your monitor, focusing on something that's at least 20 feet away. Then close your eyes for 20 seconds more. This practice also helps counter eyestrain.

Cataracts

A cataract is a clouding of the normally clear lens in the eye that occurs as fibers in the lens compress and proteins clump together. Early symptoms include blurred vision, trouble with glare, and difficulty reading even with your glasses. Eventually the opacity can become severe enough to cause a disabling loss of vision, either by distorting light rays or keeping them from reaching the retina at all.

Cataracts are extremely common among older adults. About half of all people ages 65 to 74 have them; after age 75, about 70% do. Other risk factors include obesity, diabetes, eye injuries, a family history of cataracts, and use of certain medications (particularly corticosteroids such as prednisone). Several studies have linked cataracts with alcohol consumption and smoking. But even if you have smoked for many years, quitting will help lower the chances of cataracts forming in the future. Long-term sun exposure is yet another factor; wearing sunglasses can help protect eyesight and may minimize cataract formation.

A comprehensive dilated eye exam enables an eye doctor to diagnose cataracts. The doctor will painlessly examine the interior of your eyes with a slit lamp to assess how extensive the cloudy patches are. Additional examinations and tests help rule out other eye disorders.

When cataracts first develop, you may be able to take simple steps to counter their effect. These include making sure you have enough light to see while you work and reducing glare by positioning a light directly behind you so that it shines on your work. If you wear prescription eyeglasses, get lenses with anti-reflective coating. These strategies may delay the need for surgery and perhaps allow you to avoid it entirely.

Treating cataracts

The only effective cure for cataracts is removing the clouded natural lens and replacing it with an artificial one. Cataract surgery is the most common type of eye operation performed in the United States; it is considered one of the safest of all surgeries.

The procedure is performed on an outpatient basis. It requires local anesthesia (given directly on the eye or by injection) and usually lasts less than half an hour. Most people don't feel any pain during the procedure. Afterward, the surgeon may cover your eye with a bandage or shield, which may be removed later that day or the following day.

More than 98% of people who undergo cataract surgery have improved vision afterward, assuming they have no other vision-limiting condition, and most have an uneventful recuperation. Complications are extremely rare, occurring only once in several thousand operations.

Hearing impairment

Hearing loss usually comes on so gradually, over so many years, that you may not realize that you don't hear as well as you used to. Moreover, the term "hearing loss" is misleading. It implies trouble hearing all sounds, when in fact most people start off having trouble hearing just high-pitched sounds. Initial signs of trouble may include difficulty hearing over the phone or when there's a lot of background noise. As years pass, you may find yourself increasingly asking others to repeat themselves.

Hearing loss becomes more frequent with age. About one-third of Americans ages 65 to 74 have hearing loss, while almost half of those 75 and older do. Problems with hearing can develop for many reasons, including your family history, noise exposure, infections, and chronic diseases such as diabetes, heart disease, and chronic kidney disease.

Diagnosis is usually a multistep process that can involve several health practitioners. If you notice you are having problems with your hearing, talk with your doctor to see if you need further evaluation. Your physician may refer you to a specialist. Otolaryngologists focus on problems of the ears as well as the nose and throat and are therefore commonly called "ear, nose, and throat doctors." Otologists deal exclusively with ear problems. Your doctor or one of these specialists may refer you to an audiologist, someone who specializes in testing for hearing loss and fitting hearing aids.

Managing hearing loss

Depending on your test results and your medical history, your doctor or audiologist may recommend any of several solutions. If your hearing loss is the result of a middle-ear infection, you may need antibiotics. If it's a side effect of a medication, switching to another drug may help. If your hearing loss is severe to profound, you may need a device surgically implanted in your ear to help you hear better. Still, for the majority of adults whose hearing loss is age-related or otherwise permanent, the only way to improve hearing is by wearing hearing aids.

There are many types of hearing aids available now. They are smaller than in the past, with some so small that other people don't notice them. They also

are better than they used to be at homing in on the sounds you need to hear and filtering out noise.

If your hearing loss is mild to moderate, you might consider over-the-counter hearing aids, which came on the market in 2022. These devices cost far less than custom-made hearing aids fitted by an audiologist. (Expect to pay, on average, \$1,600 for OTC devices versus \$4,600 for prescription models.) Medicare and most other insurance plans don't cover hearing aids. If you have severe hearing loss, however, you'll likely benefit more

from custom-made hearing aids from an audiologist.

It may take some time to get used to your new hearing aid—the sound will not be the same as that filtered by a healthy ear—but matching the device to your needs can help. Sorting through the options can be tricky, as there are a variety of styles (behind the ear, in the canal, etc.) and technologic features (digital, programmable, etc.). Another Harvard Special Health Report, *Coping with Hearing Loss*, delves into detail on the many hearing aid options (see “Resources,” page 53).

Musculoskeletal conditions

Bone loss

Ounce for ounce, your bones bear as much weight as reinforced concrete. But unlike concrete, bone isn't inert. It is a living tissue that the body continues to “remodel” throughout your life, depositing and withdrawing minerals (especially calcium) as needed. This process requires breaking down and rebuilding part of the bone matrix in order to release or absorb minerals.

Until age 20, the body builds new bone more quickly than it removes old bone. For the next 10 years or so, the pace slows but still favors growth. After that, bone mass in women usually remains steady until menopause, when it declines rapidly. In some women, bone loss begins in the years just preceding menopause (perimenopause), as estrogen levels start to dip.

At its extreme, bone loss can lead to osteoporosis—literally, “porous bone.” Women are more likely than men to develop osteoporosis because they have smaller skeletons; their bone loss begins earlier and occurs more rapidly; and, even at peak levels of bone density, their bones are not as densely packed with minerals as men's. But osteoporosis is not inevitable.

Preventing further bone loss

While it's true that all people lose bone as they age, there are many things you can do to hold on to the bone you have and perhaps even replace a small amount of lost bone.

Quit smoking. Tobacco use has been linked to lower bone density, and it's clear that smoking

increases fracture risk. The exact reasons are hard to pin down, as smokers also tend to be thin, drink alcohol, have poor diets, and enter menopause earlier, all of which increase the risk of osteoporosis. No matter what, quitting smoking will improve your health.

Eat a balanced diet. If you don't supply your body with the calcium it needs, it will raid the calcium stores in your bones. However, mounting evidence indicates that a high calcium intake has little effect on bone density and may increase the risk of kidney stones, so beware of supplements and fortified foods that may deliver too much calcium. The longstanding recommendation for women over 50 is to take 1,200 mg per day of calcium, half of which should come from food. One or two servings of dairy products a day can do the trick. Other good sources are leafy greens and almonds. Some types of tofu made with calcium sulfate are also calcium-rich; check the Nutrition Facts label.

Adequate vitamin D, such as 800 IU a day, is also a good idea, since it helps your body absorb calcium. The government recommends 600 IU a day for women ages 51 to 70, and 800 IU for women over age 71. As with calcium, it's best to get vitamin D from the food you eat—such as salmon, tuna, sardines, or fortified dairy products—but most people find it hard to get enough from these sources. Ask your doctor about a supplement. Modest sun exposure (five to 10 minutes on most days) on the arms and legs can also help your body make its own vitamin D.

Skin changes

Aging also affects your body's largest organ—your skin. However, the single biggest cause of damage to skin over time is not the passage of time but sun exposure. Over the years, sun exposure causes fine and coarse wrinkles; baggy skin with a yellow, leathery appearance; and pigmented areas known as lentigines or age spots. And because sun exposure diminishes collagen, which supports a network of blood vessels, photoaging can also cause skin to bruise more easily.

The most serious effect of sun exposure, though, is that it increases the risk for skin cancer. In fact, ultraviolet radiation from the sun is the most important cause of skin cancer and of the lesions that are the precursors to many skin cancers (see "Melanoma," page 34). Another Special Health Report from Harvard, *Skin Care and Repair*, covers a broad range of skin problems, from cosmetic issues to medical conditions, along with the burgeoning array of treatments that are available (see "Resources," page 53).

In addition, stick to these daily limits:

- No more than three cups of coffee a day. High amounts of caffeine may increase calcium excretion in the kidneys.
- No more than 700 micrograms (2,300 IU) of vitamin A per day. There is some evidence that higher amounts of vitamin A stimulate the production of osteoclasts, the cells that break down bone.
- No more than one drink per day. Research on alcohol and bone health is mixed, but heavy drinking seems to sap calcium from bones and alter bone-building hormones, including estrogen and cortisol. It also interferes with production of vitamin D. Some studies have linked low to moderate consumption of antioxidant-rich red wine to a lower risk of hip fracture, but even small amounts of alcohol can impair coordination and reaction time, increasing the risk of falls and related fractures.
- As little soda as possible. Although there is limited data on the effects of soda intake and bone health in older adults, research has linked soda to decreased bone density in young, healthy adults. And some studies have found that, in older adults, fracture risk started to increase for individuals who drank one to two sodas a day. Dark colored colas, including the diet variety, may be especially problematic.

Do weight-bearing exercise. Any exercise that involves supporting your body weight and working against gravity—such as running, walking, weight lifting, or stair climbing—can potentially build bone. Activities like swimming and bicycling are great for other aspects of your health, but they don't help strengthen bones because they aren't weight-bearing. Aim to get at least 30 minutes of weight-bearing exercise three to four times a week.

Get a bone density screening. As menopause recedes into the rear-view mirror, it's important to test your bone density. The World Health Organization (WHO) has developed criteria for diagnosing osteoporosis based on a measurement of bone mineral density called a T-score. These scores are most frequently determined by dual energy x-ray absorptiometry (DEXA). A T-score of -2.5 or lower marks the threshold for osteoporosis.

Most experts recommend routine DEXA screening for all women ages 65 and over. Younger women at high risk of bone loss should also be screened—namely, postmenopausal women ages 45 to 64 who weigh 132 pounds or less, and women ages 55 to 64 who weigh 154 or less and are not taking estrogen.

Your doctor may also use FRAX, a risk-assessment tool that incorporates factors besides bone density. The tool, which is available online (<https://www.fraxplus.org>), provides an estimate of the likelihood that you will fracture a hip or have another major fracture in the next 10 years. National Osteoporosis Foundation guidelines suggest that clinicians use FRAX and consider treatment with an FDA-approved drug for women who meet any of the following criteria:

- a history of hip or spine fracture
- a T-score of less than -2.5 at the hip or spine
- a T-score of -1.0 to -2.5 at the hip or spine (indicating osteopenia, which is low bone mass short of osteoporosis) together with a 10-year, FRAX-estimated risk of more than 20% for any major osteoporosis-related fracture or more than 3% for hip fracture.

If you fall into one of those categories, exercise and a diet rich in calcium and vitamin D alone won't

provide sufficient bone protection. You may also need to take a bone-building medication. Many women worry about possible side effects from osteoporosis drugs, including fractures in the upper thighbone. Although these fractures are very rare, an expert task force concluded that they were a possible consequence of long-term (five years or more) use of drugs known as bisphosphonates, including alendronate (Fosamax), ibandronate, and risedronate (Actonel). Clinicians now often recommend that women who've used these medications for a long time take a "drug holiday," stopping the drug for at least a while. If you and your doctor do decide that you should stop, be sure to have your bone density tested after a year or two. If it has declined significantly, you can always resume medication. If you're taking bisphosphonates, your doctor should periodically re-evaluate your need for continued therapy. You may need a repeat DEXA screening.

For women who are not good candidates for bisphosphonates, a medication called raloxifene (Evista) may be a good option. Raloxifene is a selective estrogen receptor modulator (SERM) used in the prevention and treatment of osteoporosis in postmenopausal women. It mimics estrogen's beneficial effects on bone density, reducing the risk of vertebral fractures without significantly affecting the breast or uterine tissues. The primary benefit of raloxifene is its ability to increase bone mineral density and decrease the incidence of vertebral fractures, thereby enhancing overall bone health. However, it also carries risks, such as an increased likelihood of deep vein thrombosis (DVT) and pulmonary embolism, similar to hormone replacement therapy. Patients may also experience hot flashes and leg cramps. The mechanism of action involves binding to estrogen receptors, thereby exerting estrogenic effects on bone and lipid metabolism while acting as an antagonist on breast and uterine tissues.

Foot problems

Simply getting older puts women at risk for foot problems. With age, the fatty pads on the bottoms of your feet become thinner and less able to absorb shock. Your feet get longer and wider because of age-related

stretching of ligaments and tendons. Wearing shoes with high heels and narrow toes is also a major contributor to pain. As a result, women have four times as many foot problems as men and account for 90% of surgical procedures for common foot ailments. The following sections explore some common foot issues.

Bunions

A bunion is a bump of bone or tissue that forms at the base of the big toe—or, less frequently, at the base of the little toe. It can rub against the inside of a shoe, inflaming the bursa (the fluid-filled sac surrounding the metatarsal joint) and causing considerable pain.

Conservative treatment. Warm soaks or ice packs, along with nonsteroidal anti-inflammatory drugs (NSAIDs; see "Oral drugs," page 16), should relieve pain. When you're on the go, protect and cushion the bunion with a moleskin pad or a semisoft orthotic shoe insert. Wearing shoes with a roomy toe box is essential.

Surgery. A bunionectomy (bunion removal, sometimes accompanied by cutting and repositioning foot bones), is often successful. Recovery may take six months and require wearing a cast or special shoes.

Plantar fasciitis

If you have pain at the bottom of your foot just in front of the heel, you may have inflammation of the



Foot pain is common in women. Choosing shoes with a lower heel and a wider toe box may help.

plantar fascia, a band of ligaments and tendons that stretches from the heel bone to the bones of the toes. The fascia forms the arch of the foot and is a shock absorber for the body. Plantar fasciitis can result from an uneven stride, poorly designed shoes, or repeated overuse during high-impact exercise.

Conservative treatment. The best therapy for plantar fasciitis is rest and cushioning the heel, either by wearing a good walking or running shoe or by using orthotics. Taking NSAIDs will reduce inflammation. After the pain has begun to subside, you should stretch the foot gently through low-impact exercise such as swimming and bicycling.

Aggressive treatment. If your pain is severe and unrelenting, your doctor may suggest steroid injections. Orthotripsy, an outpatient procedure in which shock waves are directed at the affected heel, may also be effective. Surgery to release the fascia is a last resort.

Hammertoes

This deformity of the joints causes toes to bend up and curl under like a claw. Sometimes it's caused by heredity, arthritis, or a too-tight tendon, but more often it's the consequence of ill-fitting shoes. Wearing narrow shoes with a tight toe box aggravates the condition, forcing the toe to bump up against the upper part of the shoe.

Conservative treatment. As long as the hammertoe is flexible, roomier shoes and a toe pad to keep the toe from rubbing on the shoe should relieve some of the pain. Exercises, splints, and orthotics will help position the toe more comfortably.

Surgery. If the hammertoe has become stiff, it's harder to treat. At that point, you may require an in-office procedure under local anesthesia to remove a small piece of bone so that the toe can return to a normal position. Recovery takes one to four weeks and usually requires wearing a splint and a surgical shoe.

Neuroma

Nerve irritation can produce a burning sensation or numbness in the foot and may be responsible for the development of a painful lump or thickening of tissue. In women, neuroma is due mostly to wearing narrow shoes with high heels or a constricting vamp

(the section of shoe upper that covers the front of the foot). Arthritis and repetitive stress can also cause the condition.

Conservative treatment. Neuromas can be treated with massage and by wearing roomier shoes fitted with protective pads or orthotics. Cortisone injections can also relieve pain.

Surgery. A neurectomy (surgical removal of all or part of the nerve) should relieve the pain, but it may leave a portion of the toes without feeling. Recovery usually takes two to three months.

Osteoarthritis

Osteoarthritis, the most common type of arthritis, is called a degenerative disease because it involves the deterioration of the cartilage that cushions joints (see Figure 3, page 16). Osteoarthritis is more common in women than men, particularly after age 50. Women tend to develop osteoarthritis in the knees, hips, and fingers and are also more likely to develop Heberden's nodes—hard, bony growths that form on the joint nearest the fingertip. Some postmenopausal women experience arthritis in their thumbs. Osteoarthritis of the spine is also common.

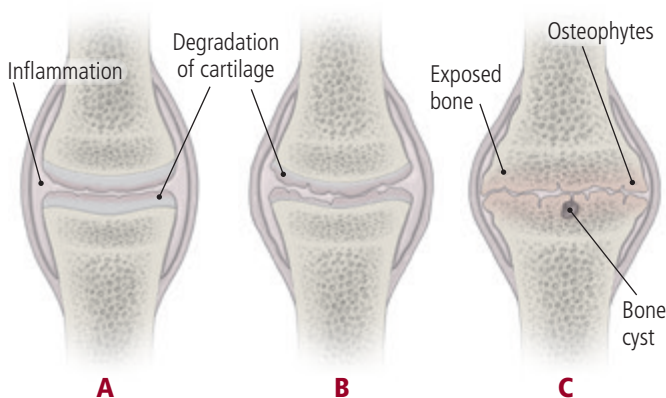
Osteoarthritis can result from genetics and also from trauma or repeated stress. In the knees, osteoarthritis may be the result of years of skiing, tennis, or another sport that torques the knee joints. Women with this problem may find it increasingly difficult to walk, climb stairs, or get up from a chair.

Some people with osteoarthritis of the hand may find that with age, their hands thicken and become stiff. Stiffness is gradually followed by pain or instability. In others, pain and stiffness subsides over time, leaving only enlarged joints as reminders.

Living well with arthritis

Arthritis can be disruptive and disconcerting. The pain and stiffness can make it difficult to perform the daily tasks most people take for granted. Simple activities like dressing yourself or cooking dinner can become a major effort. Osteoarthritis may discourage you from being active, increasing risk for other diseases, such as heart problems and obesity. The fol-

Figure 3: Joint changes in osteoarthritis



- A.** The first signs of osteoarthritis are microscopic pits and fissures on the cartilage surface, which are sometimes accompanied by inflammation.
- B.** The contours of the joint change, and the cartilage thins.
- C.** The bone surface thickens and osteophytes (abnormal bony growths) develop over time. Cartilage continues to wear away. The joint space narrows until it nearly disappears, leaving bone rubbing against bone. Bone cysts (open spaces filled with fluid) can develop as a reaction to stress on the bone.

lowing strategies and therapies can make coping with arthritis a little easier.

Regular exercise. It may seem counterintuitive, but physical activity helps maintain joint function, relieve stiffness, and decreases pain and fatigue. Exercise can also help to increase range of motion, strengthen muscles, and build endurance. As little as an hour or two of moderate physical activity each week should start to produce results. Work with your clinician or physical therapist to develop an exercise routine suited to your needs.

Achieving and maintaining a normal body weight. Bearing excess weight results in more stress on joints and more pain.

Assistive devices. A wide variety of splints for hands and fingers, wrists, elbows, knees, and ankles are available over the counter or by prescription. These may help to ease your pain and increase mobility. Using a cane can relieve pressure on hips and knees substantially. An increasing array of helpful gadgets, from jar openers to dusters to garden pruners, are also available online and at many home and hardware stores.

Acupuncture. A 2020 study that analyzed 16 review articles and 11 randomized controlled trials involving adults with chronic musculoskeletal pain conditions found that acupuncture provided short-term pain relief for people with knee osteoarthritis and chronic low back pain. If you are considering acupuncture, talk to your doctor first, and be sure to find a licensed practitioner. Keep in mind that acupuncture typically isn't covered by Medicare and may not be covered by your insurer either.

Heat therapy. A warm shower or bath can work wonders anytime, especially before and after exercising. Hot packs and moist or dry heating pads, or even a folded towel warmed in the oven or microwave, can offer instant relief.

Cooling. Gel-filled cold packs, coolant sprays, ice chips in a plastic bag, and packages of frozen peas can soothe hot, painful joints. To avoid disrupting circulation, don't leave them on longer than 20 minutes.

Drug treatment for osteoarthritis

Medication works best when paired with other pain relief strategies, such as exercising to build your muscles and protect your joints from injury. Options include the following:

Topical pain relievers. Applied to the skin, these provide mild pain relief. Creams containing salicylate (Aspercreme, Bengay) and others containing capsaicin (Capsaicin, Zostrix) are available over the counter. Since 2020, a formerly prescription-strength gel containing diclofenac (Voltaren Arthritis Pain) has also been available over the counter. Lidocaine, an anesthetic, is available in a patch by prescription and, in a slightly lower dose, over the counter. The prescription patches aren't typically covered by insurance when prescribed for osteoarthritis.

Oral drugs. Doctors often recommend acetaminophen (Tylenol) first because it's effective for mild pain and easy on the stomach. However, NSAIDs are generally considered more effective than acetaminophen in treating osteoarthritis because they reduce inflammation as well as pain. The arsenal of NSAIDs includes over-the-counter medications such as aspirin, ibuprofen (Advil, Motrin), and naproxen (Aleve), as well as the prescription drugs diclofenac, indomethacin (Indocin),

meloxicam, and nabumetone. Because NSAIDs, particularly aspirin and indomethacin, inhibit the enzyme that protects the stomach lining and interfere with blood clotting, they may increase the risks of bleeding and gastric ulcers. Some of the prescription NSAIDs, particularly meloxicam and nabumetone, seem to be relatively gentler on the digestive system. Some formulations, such as Arthrotec (diclofenac and misoprostol), combine an NSAID with a medication that protects the stomach. Be aware, however, that some NSAIDs, including diclofenac, have been linked to an increased risk of heart attack or stroke, so be certain to discuss these risks with your doctor, particularly if you have cardiovascular disease or risk factors for it.

Some drugs that are not typical pain medications may also provide relief. For example, the drug gabapentin (Neurontin) is an antiseizure drug for people with epilepsy. It can also be used to treat nerve pain, and it is approved by the FDA for the type of nerve pain that can occur in some people after having shingles. Doctors also prescribe it for other chronic pain conditions, such as osteoarthritis, although these uses are considered off-label, meaning the drug isn't specifically approved by the FDA for these conditions. Another drug that may be used for osteoarthritis is the antidepressant duloxetine (Cymbalta). This drug is FDA-approved to treat chronic musculoskeletal pain, including pain caused by osteoarthritis.

Injections. In some instances, your doctor may recommend corticosteroid injections to lessen pain and allow you to get moving again. An injection introduces a corticosteroid drug and local anesthetic into the joint to temporarily relieve osteoarthritis-related pain and swelling. But it isn't a long-term fix. There is some evidence that when cortisone injections are overused, they accelerate osteoarthritis and make the condition worse, so they should be used judiciously.

Another type of injection therapy uses a synthetic version of a hyaluronic acid, which is a component of synovial fluid—the substance that lubricates joints. Over time, the hyaluronic acid naturally found in the body can break down, leading to pain and stiffness. Synthetic forms of hyaluronic acid (Hyalgan, Synvisc) can be injected directly into an osteoarthritic joint,

theoretically restoring the lubricating qualities of synovial fluid. This treatment is approved by the FDA for knee osteoarthritis, but the jury is still out on how effective it is. Studies have yielded mixed results.

Surgery for osteoarthritis

Surgical options are usually recommended when drug therapies and other strategies are no longer effective, but before mobility is seriously limited.

Arthroscopy. Usually reserved for the knees, arthroscopy is considered minor surgery because the incisions are small and the procedure generally does not require an overnight stay in the hospital. An arthroscope (an instrument with a tiny light and a camera) and a variety of miniaturized surgical instruments are inserted into the joint through different small incisions. If there is torn cartilage to repair or debris to remove, the procedure can result in mild to moderate improvement that may last several months or perhaps a few years. It may be helpful if other treatments have failed, but recent studies have questioned its usefulness for most cases of osteoarthritis.

Joint reconstruction or replacement. Surgery can be used to correct joint deformity, to reconstruct a joint, or to completely replace a diseased joint with a prosthetic device. Long-term data from knee and hip replacements indicate that a prosthetic joint will last an average of 10 to 20 years and should relieve your pain substantially. Thanks to improved materials, newer prostheses may last even longer. If you're considering hip or knee replacement, expect to participate in physical therapy after surgery.

Alternatives to joint replacement. While joint replacement is considered the definitive treatment for a badly damaged knee or hip, it is generally a last resort and is not right for everyone. Other options include the following:

- **Realignment.** Surgery can be performed to realign bones that are no longer correctly lined up as a result of osteoarthritis.
- **Fusion.** Another possibility involves permanently fusing two bones at the affected joint. This option, although not as common, may be appropriate when arthritis occurs in parts of the body such as the wrist, ankle, and small joints of the fingers and

toes; joint replacement is less reliable in these areas and therefore rarely performed.

- **Hip resurfacing.** With hip replacement, a surgeon removes the head of the femur and replaces it with an artificial ball. With hip resurfacing, the

surgeon instead reshapes the femur head and caps it with a cobalt-chromium prosthetic that fits into a synthetic lining in the socket. Resurfacing uses a bigger ball than hip replacement does, which may make dislocation less likely.

Changes in the mind and memory

Age-related memory impairment

Many people begin to notice changes in memory by around age 50. A typical sign of this mild forgetfulness is difficulty recalling a word or name that once came easily to you. As your body ages, so does your brain. When structural changes occur, your brain loses its ability to process information quickly. Lifestyle matters, too: memory can falter as a result of stress, anxiety, fatigue, distractions, or being overloaded. Memory difficulties may also be caused by medications, poor vision or hearing, sleep disturbances, depression, or chronic pain—all things you can take steps to correct.

Cardiovascular fitness is also tied closely to brain health and memory. Any condition that compromises heart health and blood vessel flexibility can also affect memory and other mental skills. Similarly, research has linked diabetes and obesity to poorer brain health.

The good news is that many age-related memory slips are perfectly normal and not necessarily signs of dementia. There are also simple ways to help sharpen your everyday memory.

Protecting memory

Maintaining your overall cardiovascular fitness is a potent way to preserve memory function. Research also suggests that good nutrition, maintaining a healthy weight, and staying physically and mentally active in midlife can help reduce or delay memory impairment in the decades to come.

Here are some fundamental strategies to slow brain aging:

Eat a plant-based diet. Research shows that a diet that contains abundant fruits, vegetables, legumes,

nuts, and whole grains—with limited amounts of animal products—is best for cognitive fitness.

Exercise regularly. Exercise has a variety of potentially positive influences on cognitive skills. Movement enriches the blood (and therefore the brain) with oxygen. During exercise, the body produces substances that help maintain the physical integrity and function of the brain. Regular exercise also supports restful sleep, which is essential restorative “downtime” for the brain.

Get enough sleep. To learn new information, you must be able to pay attention, which you can’t do if you’re not well rested. It’s also harder to hold on to memories and retrieve information you’ve learned previously if you’re poorly rested, because the process by which you solidify recent experiences into durable memories occurs during sleep. It’s no coincidence that forgetfulness is a common symptom of insomnia.

Manage stress. Stress can be helpful in reasonable doses to motivate you and focus attention. But intense or prolonged stress takes a toll on your health. Long-term exposure to stress hormones can harm the brain—including the hippocampus, a brain region key to memory. Exercise, meditation, restful sleep, friendships and other social connections, and a positive approach to life’s challenges can all help you manage stress better and optimize memory.

Use it or lose it. Challenging your brain throughout life builds a buffer against the detrimental effects of aging on the brain. People who have led mentally active lives thanks to their education, work, and leisure activities tend to have more cognitive reserve—essentially “extra brain in the bank”—which may prevent or delay a decline in mental abilities. As you get older, you can continue challenging your brain by

learning new skills, such as taking up a musical instrument you haven't played before or studying a foreign language that's new to you.

Learn memory techniques. There are many ways to make the most of your existing memory skills. Techniques like these can help you deal with common situations that invite forgetfulness:

- Use notebooks, address books, calendars, and smart-phone apps to organize and store information for fast retrieval rather than relying entirely on memory.
- Keep items like your wallet and car keys in designated spots so you always know where to find them.
- Practice focusing more mindfully when you're taking in new information, such as when you meet new people, to enhance your recall of facts and impressions later.
- Employ memory-enhancing tricks, like making up a funny rhyme to remember a name or place, or repeating new information silently several times.

Depression

One of the most prevalent medical conditions in the United States, depression affects approximately one in 10 women at some point—twice the rate seen in men. We're not talking about routine blahs that come and go, but a clinical condition that requires treatment. A diagnosis of major depressive disorder is made if a person experiences at least five of the following symp-

toms (including at least one of the first two symptoms listed here):

- depressed mood most of the day, nearly every day
- loss of interest or pleasure in most or all activities
- change in appetite or weight
- trouble sleeping, or sleeping too much
- sluggish thinking and movement, or restlessness and agitation
- low energy
- feelings of worthlessness or guilt
- poor concentration
- recurrent thoughts about death or suicide (which requires immediate help, even in the absence of other symptoms).

Many factors contribute to depression. Moods arise in and are regulated by the brain. A woman's genetic makeup may make her vulnerable. There may be subtle problems in the structure of brain cells or the way that brain cells convey signals to one another. The brain also changes in response to its environment—such as relationships with loved ones, stressful life events, early experiences with caregivers, or exposures to medications, to name a few. Scientists are only beginning to understand all of these complicated interactions.

Together with the acknowledgment of this complexity has come the recognition that depression is not a sign of personal weakness; rather, it is a serious medical condition requiring professional treatment. Risk factors for depression include the following:

- a family history of depression, particularly in first-degree relatives
- a parent who has committed suicide
- a personal history of substance abuse or an eating disorder, or a family history of such disorders
- unresolved grief over a major life loss
- unresolved anger.

Why women get depressed

Although there is general agreement that numerous factors contribute to a higher rate of depression in women, the nature of those factors is a subject of ongoing debate. The commonly acknowledged ones include emotional “hardwiring,” developmental experiences, and social conditioning.

These factors are interrelated. Women may be more



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Major depression is a common yet serious condition that requires treatment. Medication and therapy often help.

responsive to signals from others and more reliant on external feedback for their sense of self. Women are also more likely than men to repress anger or aggression or to turn those feelings on themselves, rather than to vent them appropriately. Moreover, societal expectations may force a woman to subordinate her needs to those of others. All these tendencies may contribute to a loss of self-esteem—a central feature of depression.

A woman's risk of depression rises as she goes through menopause, and hormones may be partly to blame. Shifts in female hormone levels can cause mood changes at other stages of life, so it's not necessarily surprising that they can have some effect on mood during the menopausal transition. Hormone shifts are known to trigger conditions such as premenstrual dysphoric disorder (a more severe form of premenstrual syndrome, affecting mood) and postpartum depression, which occurs after childbirth. However, most often, the mood shifts seen during perimenopause and at menopause are mild. A study in *The Journal of Clinical Endocrinology and Metabolism* linked an increase in depression symptoms at perimenopause with fluctuations of two hormones, progesterone and estradiol (the most potent form of estrogen). But when it comes to major depressive disorder, the connection to female hormone changes is not as clear. Perhaps most importantly, there is no evidence to suggest that hormonal supplements treat depression.

Most women who develop significant mood issues during perimenopause have had them in the past. It's uncommon for a woman who has no history of depression or anxiety to suddenly develop a severe case of it during the menopausal transition. Other life events may also contribute to mood shifts during this period. Women in this age group sometimes face multiple sources of stress, including caring for children, dealing with aging parents, and navigating life changes, all of which may contribute to the incidence of depression and anxiety.

Treating depression

For most women, the immediate goal is to feel and function better as quickly as possible. In general, the milder the disorder, the more numerous the options for treating it. The two most common approaches are

Table 3: Anxiety disorders and their symptoms

DISORDER	SYMPTOMS
Generalized anxiety disorder	Exaggerated worry about health, safety, money, and other aspects of daily life that lasts six months or more. Often accompanied by muscle pain, fatigue, headaches, nausea, breathlessness, and insomnia.
Obsessive-compulsive disorder (OCD)	Obsessive thoughts, such as an irrational fear of contamination, accompanied by compulsive acts, such as repetitive hand washing, that are undertaken to alleviate the anxiety generated by the thoughts.
Panic disorder	Recurrent episodes of unprovoked feelings of terror or impending doom, accompanied by rapid heartbeat, sweating, dizziness, or weakness.
Phobias	Irrational fear of specific things or situations, such as spiders (arachnophobia), being in crowds (agoraphobia), or being in enclosed spaces (claustrophobia).
Post-traumatic stress disorder (PTSD)	Reliving an intense physical or emotional threat or injury (for example, childhood abuse, combat, or an earthquake) in vivid dreams, flashbacks, or tormented memories. Other symptoms include difficulty sleeping or concentrating, angry outbursts, emotional withdrawal, and a heightened startle response.
Social anxiety disorder (social phobia)	Overwhelming self-consciousness in ordinary social encounters, heightened by a sense of being watched and judged by others and a fear of embarrassment.

antidepressant medications and psychotherapy. Studies have indicated that either of these approaches or a combination of both can be effective.

Drug treatment. There are many factors to consider when selecting an antidepressant. Because chemistry is an important determinant of anyone's response to antidepressants, some drugs will be more effective than others for any given person. Moreover, each antidepressant is associated with a slightly different set of side effects—for example, loss of libido, constipation, sleep disturbances, or weight gain. While your clinician should have an idea of the drugs that are likely to be effective for you, you should let him or her know which side effects you consider the most or least tolerable.

Because several weeks of treatment may be necessary for an antidepressant to become fully effective, it is important not to give up on a drug too soon. Your

clinician should be able to judge when you have given a medication an adequate trial. It is often necessary to try a few antidepressants before finding a drug that effectively relieves symptoms and whose side effects are tolerable. As a general rule, drugs alone are most successful when the depression is not exacerbated by an ongoing stressful situation.

Psychotherapy. Talk therapy is most effective in the hands of a skilled professional who is also a good fit for you. Therapy can provide insight into the specific factors that precipitated the depression; guidance in changing thoughts, feelings, and behaviors that may fuel the disorder; and aid in dealing with the condition's effects on work and relationships.

However, even the most gifted therapist is not a magician who can dispel gloom immediately. Several weeks may elapse before the sessions produce a noticeable change in mood, so try to be patient.

Electroconvulsive therapy (ECT). This procedure, in which controlled seizures are induced under general anesthesia to alter brain chemistry, is usually reserved for people with severe depression for whom antidepressants have been ineffective or who are considered to be in danger of suicide. Because it is relatively fast-acting, ECT's major role is in "jump-starting" treatment. It often produces a loss of recent memory, which usually resolves over time. It is highly effective for the properly selected patient.

Anxiety

Some 40 million adults in the United States have an anxiety disorder, and women are diagnosed with anxiety nearly twice as often as men. There are a number of anxiety disorders. What they have in common is unwarranted fear or distress that interferes with daily life (see Table 3, page 20).

Anxiety is a reaction to stress that has psychological as well as physical features. The feelings associated with anxiety are thought to arise in the amygdala, a brain region that governs many intense emotional responses. As the brain transmits these signals to the sympathetic nervous system, heart and breathing rates increase, muscles tense, and blood flow is diverted from the abdominal organs to the brain.

Some degree of anxiety is normal and even necessary. In the short term, it prepares you to confront a crisis by putting the body on alert and prompting you to take extra precautions. But its physical effects can be counterproductive, causing lightheadedness, nausea, diarrhea, and frequent urination. And when anxiety persists, it can not only interfere with your daily life but also undermine your physical health. Evidence suggests that people with anxiety disorders are at greater risk for developing a number of chronic medical conditions, including heart disease, chronic respiratory disorders, and gastrointestinal conditions. They may also have more severe symptoms, and treatment of the medical conditions may be more challenging when the person has anxiety as well.

Research on the physiology of anxiety-related illness is still young, but there's growing evidence of mutual influence between emotions and physical functioning. Yet anxiety often goes unidentified as a source of other disorders, such as substance abuse or physical addiction that can result from attempts to quell feelings of anxiety. Anxiety also plays a role in somatoform disorders—mental disorders that are characterized by physical symptoms such as pain, nausea, weakness, or dizziness that have no apparent physical cause.

Treating anxiety

When anxiety is related to a medical condition or to substance use, treatment focuses on the underlying causes. Otherwise, there are three primary approaches to treatment:

Cognitive behavioral therapy (CBT). This approach centers around developing self-awareness of triggers. The cognitive component helps you identify and avoid thoughts that generate anxiety, and the behavioral part helps you learn how to react differently to anxiety-provoking situations. It has been used successfully for generalized anxiety disorder and social phobia, among others. A specific type of CBT called exposure therapy helps you confront your fears. For example, people with obsessive-compulsive disorder characterized by excessive washing may be asked to dirty their hands and wait with a therapist for increasingly longer intervals before cleaning up.

Other types of psychotherapy. Anxiety is often triggered by a deep-seated emotional conflict or a traumatic experience. Insight-oriented psychotherapy and psychodynamic psychotherapy both explore how life events, desires, and past and current relationships affect your feelings.

Drug therapy. Medications alone are less effective than psychotherapy over the long term; they may also have unpleasant side effects and interact with

other medications. Still, they can be helpful, especially when used in combination with psychotherapy. The most commonly used types of drugs include benzodiazepines, which act quickly but become less effective as tolerance builds; antidepressants, which may take longer to become effective but can be used for longer periods; and beta blockers, which reduce the physical response of anxiety by slowing heart rate and lowering blood pressure.

Less energy and muscle power

Low energy

When was the last time you said something like “I just don’t have the energy I used to have”? The underlying reasons can be hard to discern, but low energy is a common complaint of middle-aged and older adults. Sometimes the problem is linked to a serious underlying health issue, such as depression, anemia, thyroid problems, or chronic fatigue syndrome, all of which require medical attention. But this is usually not the case. Fortunately, there is a lot you can do to boost your energy besides chugging coffee and tea.

How to boost your energy

When a lack of vim and vigor can’t be tied to a specific health condition, try exercise first. Regular exercise enhances the body’s oxygen-carrying capacity. It also supports sounder, more restful sleep—another vital source of energy and one that becomes more challenging in middle age. In addition to exercising, try good sleep hygiene—a set of habits that can help improve your sleep (see “Don’t skimp on sleep,” page 44). Limiting caffeine and establishing a regular sleep routine are good places to start.

Stress management is also important. Stress-induced emo-

tions consume large amounts of energy. Exercise is a proven stress buster, but you can also try relaxation techniques, such as deep breathing. Engaging in a fun hobby can also help lower your stress levels. So can simply getting out in nature.

Underactive thyroid

Sometimes fatigue is the result of an underactive thyroid. This tiny, butterfly-shaped gland influences virtually every organ system in the body. The hormones it secretes into the bloodstream play a vital role in regulating metabolism, the rate at which your body converts food and oxygen to energy.

Women are five to eight times more likely than men to have thyroid disease, according to the American Thyroid Association. Researchers don’t fully understand why women are more susceptible, but the difference might be linked to estrogen.

Low thyroid hormone production, or hypothyroidism, causes a range of symptoms. In addition to fatigue, you may have constipation, dry skin, brittle nails, aches and pains, increased sensitivity to cold, appetite loss, weight gain (because of slowed metabolism), memory issues, and loss of interest in things that are normally impor-



Getting regular exercise, limiting caffeine, and sticking to a regular sleep routine can help improve the quality of your sleep.

tant to you. In general, the lower your thyroid hormone levels are, the more pronounced your symptoms will be. Untreated hypothyroidism can also increase your risk for high cholesterol, high blood pressure, and heart disease.

A simple blood test can reveal whether you have abnormal levels of thyroid-stimulating hormone (TSH) and the thyroid hormone thyroxine (T4).

Treating underactive thyroid

Hypothyroidism is usually treated with a daily dose of synthetic T4 (levothyroxine sodium), taken in pill form. Levothyroxine works exactly like your body's natural thyroid hormone. It's available in generic form and under such brand names as Unithroid, Levoxyl, and Synthroid. Some people also require a small dose of synthetic T3 (Cytomel). When treatment is effective, your levels of TSH (which prompts your thyroid to produce more thyroid hormone) should normalize. Typically, you'll start with a relatively low dose and have your TSH checked six to eight weeks later. If necessary, your physician will adjust the dose, repeating this process until your TSH is in the normal range. Once the right dose is established, your TSH and possibly T4 levels will be checked once or twice a year. Most people who take enough synthetic T4 to normalize TSH levels find that their symptoms disappear.

Reduced muscle strength

Gradual loss of muscle mass is a natural part of aging. It starts in your 30s but accelerates after age 60. Over time, you perceive this as a decrease in both muscle strength (the amount of weight you can lift) and power (how quickly you can pull, push, or lift).

Preserving muscle mass is important to your overall health and quality of life. In addition to the obvious benefits, greater muscle mass is also associated with a lower risk of problems such as heart disease, diabetes, obesity, and general loss of mobility. As for power, it helps not only in sports, but also in daily life as you age—for example, by helping you cross the street quickly or preventing a trip from turning into a fall.

Staying strong

No matter how old you are, the key to recovering and keeping muscle mass and function is progressive resistance training. This means working your muscles with free weights, exercise machines, your own body weight, or elastic bands—and continually increasing the amount of resistance in manageable increments. Studies have found that as little as six weeks of progressive resistance training can dramatically improve strength and power as well as mobility and balance. Resistance training complements your overall cardiovascular fitness, in combination with aerobic exercise like walking and jogging. ♥

Reducing your risk for major illnesses and injuries

While certain changes are all but inevitable as you move beyond midlife, other, more serious problems are not necessarily part and parcel of the aging process. Admittedly, age is a risk factor for most of them, but you can take steps to reduce your likelihood of developing these conditions, or at least reduce their severity. Problems like heart disease, cancer, and chronic obstructive pulmonary disorder—which consistently rank in the top 10 causes of death—are years in the making, offering you multiple opportunities to intervene with lifestyle changes that can help you counteract years of earlier behavior and even override the influence of genes to a large extent.

This chapter is devoted to those major threats, which fall into five broad categories: cardiovascular disease, cancer, respiratory disease, Alzheimer’s disease, diabetes, and accidents. In each section, you’ll see a boxed list of specific actions you can take to lower your risk. The details of how to implement those changes are provided in the Special Section, “10 steps to a longer and healthier life,” which starts on page 42. While some changes may seem challenging at first, it’s worth the effort. Lifestyle changes are more powerful than any medicine yet invented. Just as important, following these steps will make you feel better now and in the years to come.

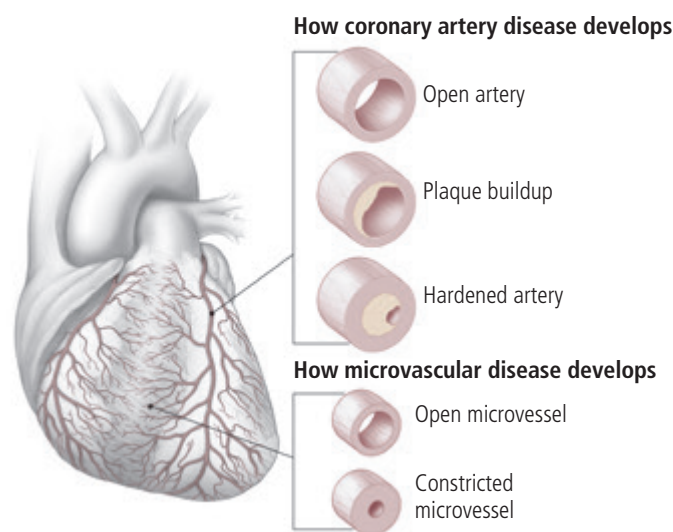
Cardiovascular disease

Heart disease

Heart disease is the umbrella term for a variety of problems that affect the heart or the blood vessels that nourish it. It is the leading cause of death in women. According to the CDC, it kills over 300,000 women per year—more than all types of cancer combined. Women typically begin to show symptoms about a decade later than men, as the protective effects of estrogen dwindle following menopause. What’s more, women sometimes present with different symptoms, which has the potential to lead to treatment delays.

The most common type of heart disease in the United States is coronary artery disease. As the name implies, it affects the blood vessels that wrap around the heart, bringing oxygenated blood to the heart muscle itself. This includes the major coronary arteries, which are on the surface of the heart, and a

Figure 4: Two causes of coronary artery obstruction



According to the classic model of heart disease, trouble begins when plaque builds up inside a large coronary artery, which eventually thickens and hardens, obstructing blood flow to the heart muscle. The newer, broader view of ischemic heart disease recognizes that tiny vessels feeding the heart can also become stiff and narrow, reducing blood flow.

What's your risk of having a heart attack?

You may already have a general idea of whether your risk is high or low, but you can put a number on it by using an online risk calculator. The American Heart Association and the American College of Cardiology offer one at www.health.harvard.edu/heartcalculator. To use it, you'll need to know your total and HDL cholesterol values and your blood pressure. It also takes into account your age, sex, race, and smoking status. After you plug in all the information, it will give you your estimated risk of having a heart attack in the next 10 years. However, calculators like this are based on the results of large population studies, meaning that individual risk factors affecting smaller groups of people may not be part of the equation. For example, this calculator does not account for early menopause or having had gestational diabetes (see "How heart disease risks differ for women," page 26). Even if the calculator does not indicate that you are at high risk, such additional risk factors might prompt your doctor to be more aggressive when it comes to preventive strategies.

web of ever-smaller vessels that branch from the coronary arteries to saturate the heart tissue with blood.

Trouble often begins when excess cholesterol—carried through the bloodstream in protein-coated particles known as LDL, or “bad” cholesterol—builds up in artery walls. In a perfect world, you would have sufficient HDL, or “good” cholesterol particles, to sop up the excess and carry it to the liver for disposal. But many people do not have enough HDL. The immune system sees the cholesterol that remains in vessel walls as an invader and launches an inflammatory attack. White blood cells gobble up the cholesterol, becoming engorged with fat and mired in the vessel walls, where many of them die, spilling their contents. In an attempt to isolate the damage, the immune system then encases the area in a fibrous coat, forming atherosclerotic plaque. But because the cholesterol hasn't been eliminated, the inflammatory process persists.

According to the classic model of coronary artery disease, this plaque can reduce blood flow to the heart, causing chest pain (angina). It can also lead to heart attacks in one of two ways. Plaque can build up inside a coronary artery until it blocks blood flow completely, starving the heart muscle of oxygen and nutrients. Or,

more frequently, unstable plaque can rupture under pressure. A blood clot then forms over the ruptured plaque, and, if the clot is large enough, it blocks the artery, cutting off blood flow.

While most heart attacks result from a clot in a coronary artery, a small number of people (5% to 10%) appear to have heart attacks even when their arteries are clear. This occurs more frequently in women than in men. They generally have the same symptoms as anyone else experiencing a heart attack, but when doctors perform an angiogram (an imaging test of the heart's arteries), they can't find a blockage. Doctors call this condition myocardial infarction with non-obstructive coronary arteries, or MINOCA for short. Experts are still trying to understand why MINOCA occurs, but they suspect there is more than one possible cause. Among them are the following:

- **Coronary artery spasm.** When blood pressure rises, an abnormality in the artery causes it to constrict and block blood flow long enough to produce heart attack symptoms. After the spasm ends, there is no blockage to be found on the angiogram.
- **Microvascular disease.** MINOCA could also be caused by blockages in tiny arteries that a doctor isn't able to see on an angiogram (see Figure 4, page 24).
- **Blood clots.** In some instances, MINOCA may occur when a tiny blood clot breaks loose and blocks an artery, but clears by the time the doctor performs the angiogram.
- **A tear in a coronary artery.** This problem, also known as spontaneous coronary artery dissection, typically occurs in young or middle-aged women and is often undetectable on angiograms.

While doctors used to think that these types of heart attacks were harmless, research shows MINOCA shouldn't be ignored. A study published in the *International Journal of Cardiology* found that a quarter of people who were diagnosed with MINOCA went on to have another major cardiovascular event, such as another heart attack or a stroke. They may also eventually develop heart failure. Prevention of MINOCA involves the same basic measures that you would take to prevent classic heart disease.

Risk factors

Many factors interact to increase your risk of developing heart disease and having a heart attack. A number of them are common to both women and men, but women have some unique risk factors, too (see “How heart disease risks differ for women,” below).

Age. After 55, a woman’s chance of having a heart attack begins to rise steadily (see “What’s your risk of having a heart attack?” on page 25). Not coincidentally, by that time, most women have gone through menopause, which means the protective effects of estrogen—raising HDL and lowering LDL cholesterol levels—have disappeared.

Family history. You have a strong genetic risk for heart disease if you have a mother or sister who was diagnosed with coronary artery disease before age 65 or a father or brother who developed it before age 55.

Ethnicity. Blacks, Mexican Americans, Native Americans, and native Hawaiians are all more likely than people of European descent to develop heart disease. It has become increasingly clear that systemic racism plays an important role in such disparities in health outcomes.

Smoking. Smoking is the leading preventable cause of death in the United States, and one in three smoking-related deaths is from cardiovascular dis-

How to reduce your risk of heart disease

- Get active (see “Keep moving,” page 42).
- Keep your cholesterol in check (see “Unfavorable cholesterol profile,” page 27).
- Improve your diet (see “Adopt a pattern of healthy eating,” page 43).
- Keep your blood pressure within healthy limits (see “High blood pressure (hypertension),” page 27).
- Watch your weight (see “Measure your waist,” page 46).
- Reduce blood sugar, if it’s high (see “Diabetes,” page 39).
- Shun cigarettes (see “Kick the smoking habit,” page 45).

ease. Even nonsmokers increase their heart disease risk by 25% to 30% when they live or work with smokers. That said, the damage is reversible. Within a year of quitting, former smokers have slashed their risk of a heart attack in half. Fifteen years out, the risk of a former smoker having a heart attack is the same as someone who doesn’t smoke.

Diabetes. An adult diagnosed with diabetes has the same high cardiac risk as someone who has already had a heart attack. Some 68% of people over age 65 with diabetes will die from some type of cardiovascular disease—a death rate that is two to four times that of

How heart disease risks differ for women

Increasingly, experts are noting risk factors for heart disease that are specific to women. If you otherwise aren’t considered at high risk but have one or more of the following risk factors, discuss it with your doctor. It might prompt your doctor to address your cardiovascular health more aggressively.

An early menopause. Research has found that women who go through menopause before age 40 have nearly double the risk of experiencing a nonfatal heart attack, stroke, or related problem before their 60th birthday when compared with women who go through menopause around age 50.

Gestational diabetes. Gestational diabetes affects nearly one in every 10 pregnant women, according to the CDC. It is most often diagnosed in the second half of a pregnancy. Women with gestational diabetes have elevated blood sugar because their bodies become resistant to insulin during pregnancy. Although gestational diabetes goes away after the baby is born, women who experience it have a higher risk of eventually developing diabetes, cardiovascular disease, or both.

Pre-eclampsia and gestational hypertension. It’s thought that pregnancy, because of the demands it places on the body, may act as a “stress test” of sorts for women, revealing underlying problems that can lead to cardiovascular disease in later years. Having either gestational hypertension (high blood pressure that develops during pregnancy) or pre-eclampsia (high blood pressure during pregnancy plus damage to another organ system, such as the liver or kidneys) can increase a woman’s risk of developing cardiovascular disease later in life. Pre-eclampsia can double or even triple the risk.

A preterm birth. A study published in the journal *Hypertension* found that women who gave birth prematurely (before the 37th week of pregnancy) were more likely to experience rising blood pressure later, compared with women who delivered closer to term. If they had both a preterm birth and a later rise in blood pressure, they were more likely to show signs of coronary artery disease.

the general population. Diabetes can also cause chronic kidney disease, which in turn can increase the risk of cardiovascular disease even more. That's because damage to the kidneys' filtering capacity allows substances like renin (which increases blood pressure) to build up in the bloodstream, harming blood vessels.

Unfavorable cholesterol profile. The more LDL in your bloodstream, the greater your heart disease risk. Experts categorize LDL levels below 100 milligrams per deciliter (mg/dL) as optimal, those 100 to 129 as near to above optimal, 130 to 159 as borderline high, 160 to 189 as high, and greater than 190 as very high.

By contrast, the more HDL in your bloodstream, the lower your chances of having a heart attack. Levels of 60 mg/dL or above are considered protective against heart disease, while levels below 40 mg/dL increase your risk.

High blood pressure (hypertension). High blood pressure damages the cells lining arteries, prompting the immune system to mount an inflammatory response that contributes to the development of atherosclerotic plaques. These plaques cause the arteries to stiffen, which drives up blood pressure even more, setting up a vicious cycle. If these plaques enlarge, rupture, or break off, they can also lead directly to heart attacks and strokes. The chance of developing complications increases with higher blood pressure levels.

Your blood pressure reading has two parts. The first and higher number (systolic blood pressure) represents the pressure while the heart is beating and shows how hard the heart works to push blood through the arteries. The second and lower number (diastolic blood pressure), which represents the pressure when the heart is relaxing and refilling with blood between beats, shows how forcefully arteries are being stretched most of the time.

The American Heart Association and the American College of Cardiology currently define blood pressure readings as follows:

- Normal blood pressure is below 120/80 millimeters of mercury (mm Hg).
- Blood pressure is considered elevated if the systolic number is 120–129 and the diastolic number is below 80.
- Stage 1 hypertension is diagnosed if the systolic num-

Where's the salt?

To keep your blood pressure in check, it may help to cut back on your salt intake. The Dietary Guidelines for Americans recommend limiting sodium to 2,300 milligrams (mg) per day for the general population, and suggest that people 51 and older who have hypertension or prehypertension should consume merely 1,500 mg. That's less than half the amount the average American consumes. Restaurant meals are often loaded with salt, as are many processed foods—including many breads, cheeses, and other products that don't necessarily taste salty.

Check the sodium content on the Nutrition Facts panel of every packaged food you buy. It will be listed both in milligrams per serving and as a percentage of the Daily Value (2,300 mg).



ber is 130–139, the diastolic number is 80–89, or both.

- Stage 2 hypertension is diagnosed if the systolic number is 140 or higher, the diastolic number is 90 or higher, or both.

Inflammation. Mounting evidence that inflammation is a major player in atherosclerosis has led researchers to develop a test for a marker of blood vessel inflammation. For years, doctors have measured blood levels of C-reactive protein (CRP), an indicator of inflammation, to monitor diseases such as pneumonia, rheumatoid arthritis, and lupus. However, a high-sensitivity test called hsCRP is more specific to blood vessels. Studies show that people with the highest CRP levels—more than 3 milligrams per liter (mg/L)—are about twice as likely to develop coronary artery disease and suffer a heart attack or other cardiac event as people with the lowest levels (below 1 mg/L). The test is most useful as a tiebreaker when considering whether a woman should take medication to reduce her overall cardiovascular risk.

Anemia. Anemia is a condition in which a person does not have enough red blood cells. These cells carry oxygen to your tissues, so anemia can leave you feeling tired or weak as it reduces the heart muscle's oxygen supply. The WISE study, funded by the National Institutes of Health, determined that cardiovascular outcomes are worse for anemic women than for

women with normal levels of hemoglobin (the molecule that carries oxygen in red blood cells). In older women, anemia commonly results from inflammation; decreased iron absorption due to a reduction in gastric acid; declining kidney function, which slows hemoglobin production; or certain medications, including cephalosporins, penicillin, and nonsteroidal anti-inflammatory drugs.

Overweight and obesity. Not only is obesity closely linked to high blood pressure, unfavorable cholesterol levels, lack of exercise, and diabetes, but it also increases your risk for heart disease independent of these other conditions. All forms of obesity are bad for health, but excessive fat around the waistline (the “apple” shape) is more dangerous to the heart than fat at the hips and thighs (the “pear” shape).

Sedentary lifestyle. Inactivity roughly doubles the risk for coronary artery disease, making it as risky as smoking, high cholesterol, or high blood pressure. By contrast, regular physical activity helps to reduce many heart risk factors. It raises HDL cholesterol levels while lowering triglycerides, blood sugar, and blood pressure. In addition, exercise helps alleviate mental stress, which can be a trigger for heart problems. Sitting for hours at a time can increase the risk of cardiovascular disease and diabetes, even for people who exercise. If you’re working on a project or job that keeps you chair-bound, it’s a good idea to stand—and even better to take a short walk—every 30 minutes or so.

Psychological factors. Your mental health can also affect your heart disease risk in a number of ways.

- **Stress.** Stress can raise blood pressure, reduce blood flow to the heart, decrease the heart’s pumping ability, trigger abnormal pumping rhythms, and activate the blood’s clotting system and its inflammatory response. Surprisingly, research indicates that chronic stress may be more harmful to your heart than major life changes. One large study found that women who cared for a disabled spouse for at least nine hours a week faced a higher chance of having a heart attack or dying from heart disease than women without such spousal responsibilities.
- **Depression.** The relationship between depression and heart disease is a two-way street. Depression

roughly doubles your risk of developing coronary artery disease, according to one review article. Other studies show that people who already have heart disease are three times as likely to be depressed as other people. As many as one in five heart attack survivors develops depression. And depression is an independent risk factor for a subsequent heart attack in people who’ve already had one. This may be in part because people who are depressed are less likely to quit smoking, take prescribed medications, or exercise—even after a heart attack.

- **Hostility and anger.** People who are habitually angry are two to three times as likely to have a heart attack or other cardiac event as others, according to one review article.
- **Social isolation.** Evidence shows that men and women who live alone are significantly more likely to have a heart attack or die suddenly from one. On the flip side, older adults with a strong network of friends are significantly less likely to die over a 10-year period than those without robust social support.

Stroke

Strokes occur when part of the brain does not get enough blood, most often as the result of a blockage in a blood vessel (which causes an ischemic stroke). You may be surprised to learn that about 60% of stroke victims in the United States are women. More women than men also die of strokes.

How to reduce your risk of stroke

- Get active (see “Keep moving,” page 42).
- Keep your cholesterol in check (see “Unfavorable cholesterol profile,” page 27).
- Improve your diet (see “Adopt a pattern of healthy eating,” page 43, and “Where’s the salt?” on page 27).
- Keep your blood pressure within healthy limits (see “High blood pressure (hypertension)” page 27).
- Watch your weight (see “Measure your waist,” page 46).
- Reduce blood sugar, if it’s high (see “Diabetes,” page 39).
- Shun cigarettes (see “Kick the smoking habit,” page 45).

How to recognize stroke symptoms

To help you spot a stroke, memorize the acronym F.A.S.T.—which stands for face, arm, speech, time:

Face. Smile. Does one side of your face droop?

Arm. Raise both arms. Does one drift down?

Speech. Try to talk. Are your words slurred or garbled?

Time. If the answer to any of the questions above is “yes,” there’s a high probability that you’re having a stroke. Time is important, so call 911 immediately. The sooner you get to the hospital, the better the chances of recovering or minimizing the damage.

A number of factors put women at higher risk than men. As estrogen levels fall at menopause, blood pressure tends to rise, and with it the risk of strokes. Pre-eclampsia (high blood pressure and organ damage during pregnancy) doubles the risk of stroke later in life. Women also have a higher rate of migraine headaches than men do. Migraines, when accompanied by a visual disturbance known as an aura, are associated with an increased risk of strokes.

Stroke is sometimes called a “brain attack.” Just as heart cells die in a heart attack when their blood supply is cut off, a disruption in blood flow to the brain kills brain cells. Damage from strokes is sometimes reversible, but strokes can also result in permanent loss of the ability to move one side of the body or the loss of speech or other functions, depending upon what area of the brain is affected. The severity of the damage depends on factors such as the location of the stroke, the extent of tissue injury, and how quickly symptoms are treated.

There are two basic types of strokes: ischemic and hemorrhagic. Ischemic strokes, the most common type, occur when a clot blocks an artery supplying blood to the brain. The clot may form in a blood vessel within the brain (thrombotic stroke), or it may form elsewhere and travel to the brain, where it lodges in a narrow vessel (embolic stroke). If the blood supply is interrupted only temporarily, so that symptoms go away in less than a day, it’s called a transient ischemic attack (TIA), sometimes referred to as a warning stroke or ministroke. A TIA must be taken seriously and treated as an emergency because, at the

start, there’s no way to distinguish it from a full-blown stroke. In addition, about one-third of those who experience a TIA will go on to have a full stroke, often within a year.

In hemorrhagic strokes, which make up 10% to 20% of cases, a blood vessel in the brain bursts. Not only are brain cells deprived of the blood supplied by the vessel, but surrounding tissue is also damaged, as leaking blood irritates brain cells and creates pressure on parts of the brain.

After a stroke, female survivors are more likely to be disabled and more likely to enter a nursing home than males are. One reason women fare worse than men is that, on average, they suffer their first stroke later in life, when they are more likely to be in declining health and less likely to have someone at home to care for them. Another reason is that women may be less likely to acknowledge the signs of a stroke and to get help as quickly as men do. When in doubt, seek medical attention.

Risk factors

Many of the factors that put you at risk for heart disease also increase the risk of stroke. However, some may be more important than others, depending on your individual situation.

Age. The risk of stroke increases with each decade of life. Since age alone is enough to elevate your stroke risk, it’s increasingly important to try to lower your other risk factors as you get older.

Family history. Having a close relative with heart disease or a history of stroke increases stroke risk.

Race. Black women have a greater risk of having a stroke, and dying from one, than white women do—in part because they often lack equal access to health care.

A previous stroke or TIA. Women who have a stroke between ages 40 and 69 have a 22% chance of having another within five years of the first. Women who have a TIA are also at greater risk of having a stroke.

High blood pressure (hypertension). Untreated high blood pressure is the leading cause of strokes. It damages the lining of blood vessels, setting the stage for both clot formation (ischemic strokes) and vessel rupture (hemorrhagic strokes).

Atrial fibrillation. This condition increases your risk of stroke about fivefold. It is a heart rhythm disturbance in which the upper chambers of the heart (the atria) periodically quiver erratically rather than beating forcefully, allowing blood to pool and, sometimes, form stroke-causing clots.

Smoking. Smoking damages the lining of blood vessels, posing a stroke risk. Menopausal women who both smoke and take oral contraceptives (which sometimes are used to regulate bleeding around the time of menopause) are at increased risk of having a stroke or other cardiovascular crisis. (Postmeno-

pausal hormone therapy, which contains lower hormone doses than oral contraceptives, does not pose as much risk.)

Unfavorable cholesterol profile. Low HDL cholesterol levels (below 40 mg/dL) are a stronger risk factor for women than for men.

Physical inactivity. Lack of exercise, which is directly linked to an increased risk of heart disease, contributes to obesity and other risk factors for stroke.

Obesity and overweight. Being overweight increases the risk of strokes and heart disease, in part because it increases low-grade inflammation.

Cancer

Breast cancer

The statistical picture of breast cancer has brightened, thanks to advances in diagnosis and treatment. More tumors are being caught at an early stage, and the death rate has dropped. You can estimate your risk for breast cancer by using the National Cancer Institute's online calculator at www.cancer.gov/bcrisktool.

Risk factors

Some breast cancer risk factors are within your control, while others aren't, but educating yourself and changing what you can may tip the odds in your favor.

Family history. Only 5% to 10% of all breast cancer cases are hereditary, meaning they are linked to a BRCA1 or BRCA2 gene mutation or another less common mutation. Women of Ashkenazi Jewish heritage are about five times more likely than other women to have a BRCA1 or BRCA2 mutation. If you have genetic testing and are negative, you can still develop breast cancer and should continue with screening. If you test positive, however, it means your risk is very high—you have a 45% to 85% chance of getting breast cancer by age 70. In that case, talk to your doctor about risk-reduction options. You might consider getting screened more often, using chemoprevention (taking medication such as tamoxifen or raloxifene), or having a bilateral mastectomy, oopho-

rectomy, or both. Ovary removal reduces the risk of ovarian cancer, which is also linked to BRCA1 and BRCA2 mutations.

Estrogen exposure. Because estrogen fuels breast cell growth, the longer your ovaries were producing estrogen, the greater your risk of breast cancer. Any of these conditions means greater estrogen exposure: menstruation before age 12, menopause after 55, pregnancy after 35, or no pregnancies. Postmenopausal hormone therapy adds to the risk.

Breast history. Dense breasts, which have more fibrous or glandular tissue and less fat, may increase your risk, as they can indicate a propensity for tissue to proliferate. Prior breast biopsies, especially those

How to reduce your risk of breast cancer

- Improve your diet (see "Adopt a pattern of healthy eating," page 43).
- Watch your weight (see "Measure your waist," page 46).
- Stay active (see "Keep moving," page 42).
- Follow mammography screening advice (see "When do you need a mammogram?" on page 31).
- Minimize your exposure to estrogen (see "Estrogen exposure," above).
- Avoid alcohol, but if you do drink, have no more than one drink per day (see "Rethink your drink," page 45).

When do you need a mammogram?

In 2024, the U.S. Preventive Services Task Force (USPSTF) updated their breast cancer screening guidelines. They now recommend that all women get screened for breast cancer every other year starting at age 40 and continue through age 74. Previously they recommended that women age 40 to 50 make an individual decision with their clinician based on their personal risks. Similarly, guidance from the American Cancer Society (ACS) states that women between age 40 and 44 should have the option to start screening with a mammogram every year and that all women age 45 to 54 continue with yearly mammogram screening. The table below provides a summary of both sets of recommendations.

Table 4: Mammography guidelines for average-risk women based on age

AGE	ACS GUIDELINES	USPSTF GUIDELINES
40 to 44	Option to start annual screening	Screening every other year
45 to 49	Annual screening	Screening every other year
50 to 54	Annual screening	Screening every other year
55 and older	Option to continue annual screening or screening every other year as long as a woman is healthy and has a life expectancy of at least 10 years	Screening every other year through age 74; insufficient evidence to recommend for or against screening for women 75 and older

Sources: American Cancer Society; U.S. Preventive Services Task Force.

The current gold standard is 3D mammography, which is also known as tomosynthesis. The machine takes many low-dose x-rays at multiple angles as it moves over the breast and combines

them into a 3D picture that allows a radiologist to see whether a tumor is present.

Although not part of the new USPSTF guidelines, some other breast-imaging

tests are helpful for certain patients:



Ultrasound. If your doctor feels a lump in your breast or if a mammogram detects one, an ultrasound might be suggested to determine if the lump is solid (which could be cancer) or fluid-filled (which is usually a benign cyst). Ultrasound may also be used in addition to mammography for some women with dense breast tissue.

Magnetic resonance imaging. This test is recommended for screening women at high risk (greater than 20% lifetime risk) for breast cancer, such as those with a BRCA gene mutation, those who have received radiation for Hodgkin's lymphoma, and those who are at high risk based on a combination of risk factors (such as family history, dense breasts, or onset of menstruation before age 12).

Because there is no evidence that breast self-exams save lives, neither the ACS or the USPSTF guidelines advise women to do them. Instead, practice "breast self-awareness"—which means knowing what your normal breasts are like so you can alert your doctor to changes.

that revealed precancerous abnormalities such as lobular carcinoma in situ, are also linked to a higher risk. And having had one cancerous breast tumor increases the risk of another malignancy.

Radiation therapy. Chest radiation to treat a cancer such as lymphoma increases the risk of developing breast cancer later in life, especially if you had the radiation during puberty. In women who have a BRCA gene mutation, the risk is amplified, especially for chest radiation received before age 20.

Obesity. Obesity raises the risk of breast cancer, especially after menopause. This is probably because body fat produces some estrogen.

Alcohol use. Alcohol also increases circulating

estrogen levels, so breast cancer risk rises with every drink taken. The American Cancer Society now recommends avoiding alcohol entirely to reduce the risk of various cancers, not just breast cancer.

Lack of exercise. Growing evidence is implicating inactivity as a contributing factor to breast cancer.

Lung cancer

As rates of smoking have declined, so have the numbers of new lung cancer cases and deaths. While that's good news, lung cancer remains the leading cause of cancer death in women. What's more, the disease is on the rise among nonsmokers—especially female nonsmokers.

How to reduce your risk of lung cancer

- Shun cigarettes (see “Kick the smoking habit,” page 45).
- Test your house for radon and take steps to reduce it if necessary.
- Avoid exposure to environmental toxins if at all possible.

While overall rates of lung cancer are higher in men versus women, a 2023 report from the American Cancer Society shows that women age 35 to 54 are now *more* likely than their male peers to be diagnosed. The reason why is not clear, but a 2024 review published in *Clinical Lung Cancer* notes that biological differences likely play a role.

Risk factors

Smoking still leads the list of risk factors for lung cancer, but it is not the only one.

Tobacco smoke. More than 84% of women with lung cancer are present or former smokers. However, passive smoking also makes you susceptible; living with a smoker can increase your risk of lung cancer by 20% to 30%.

Radon. Exposure to the radioactive gas emitted by uranium as it decays is the second leading cause of lung cancer deaths. Living in homes with radon levels above 4 picocuries/liter—the level deemed safe by the Environmental Protection Agency (EPA)—may increase the risk for lung cancer. Radon is odorless and colorless. You can find information on testing

your home and order test kits from the EPA website at www.epa.gov/radon.

Environmental toxins. Asbestos, organic chemicals, tar, soot, and air pollution have been implicated in lung cancer development, but the relative risk posed by each of these substances hasn’t been established.

Genetics. People with a first-degree relative who developed lung cancer, especially at an early age, may be at elevated risk.

Colorectal cancer

The second most common cancer found in women—and the third leading cause of cancer deaths in both sexes—is colorectal cancer. This is an umbrella term covering both colon cancer and rectal cancer, but the majority of these cancers occur in the colon. In both locations, they tend to arise from growths called adenomatous polyps on the inner lining of the organ.

Because the progression from precancerous polyp to full-blown cancer generally takes 10 years or more, regular screening can literally be a lifesaver. The most thorough screening method for both colon cancer and rectal cancer is colonoscopy, which allows physicians to remove polyps before they become malignant and to identify cancer early, when it’s most treatable. Colorectal cancer is highly treatable when caught early.

During a colonoscopy, a gastroenterologist examines the lining of the colon and rectum with a lighted scope, looking for polyps and cancerous lesions. There are other screening methods—testing that looks for hidden blood in stool, flexible sigmoidos-

Finding lung cancer early

In 2021, the U.S. Preventive Services Task Force (USPSTF) published new screening guidelines for lung cancer. The guidelines recommend annual screening with low-dose computed tomography (CT) if *all* of the following apply:

- You’re 50 to 80 years old.
- You have smoked the equivalent of a pack a day for a 20-year period.
- You are either still smoking or you quit within the past 15 years.

Screening can stop when you have gone 15 years without

smoking or you develop a health problem that substantially limits your life expectancy or makes you unwilling or unable to undergo treatment if you do have cancer.

CT imaging has the potential to find lung tumors early, when they’re more likely to be successfully treated. But screening CT scans aren’t recommended for everyone, because they often detect abnormalities that aren’t cancer but must still be assessed with additional scans, biopsies, or surgeries, which carry their own risks. They also expose you to some radiation, albeit a low dose.



Eating a lot of red meat might increase the risk of colorectal cancer.

copy, CT colonography (also called virtual colonoscopy), and a stool test for DNA shed by colorectal cancers called sDNA-FIT. However, traditional colonoscopy is best because it checks the entire colon, detects the most polyps and lesions, and allows

the doctor to biopsy or remove worrisome growths on the spot. Colonoscopy is particularly important for women, since they are more likely to have polyps or lesions in the ascending colon, the area farthest from the colonoscope. Sigmoidoscopy examines only the first third of the colon.

If you are age 45 to 75, you should be having regular colorectal cancer screenings. There are several screening tests available:

- a stool-based fecal occult blood or fecal immunochemical test (FIT) every year
- a stool sDNA-FIT test every 1 to 3 years
- flexible sigmoidoscopy every 5 years, or every 10 years with FIT performed every year
- CT colonography (virtual colonoscopy) every 5 years
- colonoscopy every 10 years.

If you are under age 45, talk to your provider about getting screened. Early screening may be recommended if you have a strong family history of colon cancer or polyps, or if you have inflammatory bowel disease or a history of polyps.

Risk factors

Anyone can get colorectal cancer, but certain people are more vulnerable.

Age. The vast majority of colorectal cancers occur in people over age 50, and the average age of diagnosis in women is 72.

Family history. Having at least one first-degree relative with colorectal cancer doubles the risk of devel-

How to reduce your risk of colorectal cancer

- Undergo periodic screening for colorectal cancer (see Table 5, page 47).
- Limit consumption of red meat and processed meats (see "Adopt a pattern of healthy eating," page 43).
- Shun cigarettes (see "Kick the smoking habit," page 45).

oping it yourself. Two rare, inherited syndromes also raise your risk. One, called familial adenomatous polyposis, accounts for just 1% of all colorectal cancers—but nearly everyone who carries genetic mutations for it develops cancer by age 40 unless they have preventive surgery to remove the colon. The other, hereditary non-polyposis colorectal cancer (also known as Lynch syndrome), increases lifetime risk of colorectal cancer as much 80%, although fewer than 5% of all colon cancers result from this syndrome.

Personal history of polyps. Although most polyps do not develop into cancer, people who have had polyps removed in a colonoscopy are at a higher risk and may require more frequent colonoscopies than people at average risk.

Inflammatory bowel disease (IBD). Having ulcerative colitis or Crohn's disease increases the risk up to five times; however, more than 90% of people with IBD do not get colon cancer.

Diet. Research suggests that people who consume a lot of red meat have higher colorectal cancer rates compared with those whose diets contain little or no red meat. Potentially worse are cured and processed meats (such as hot dogs, sausage, salami, ham, bacon, and jerky). The World Health Organization classifies cured and processed meats as a Group 1 carcinogen, meaning that there is sufficient evidence to state that they raise the risk of cancer, in this case colon cancer—though the overall risk these foods pose is low compared with, say, the risk that smoking poses for lung cancer.

Sedentary lifestyle. Several studies have linked a lack of exercise and sedentary behavior to a higher risk of colorectal cancer.

How to reduce your risk of ovarian cancer

- Eat a healthy diet and stay active.
- If you're high risk, consider preventive surgery to remove the fallopian tubes and ovaries.

Ovarian cancer

Although it is not in the top 10 list of the most common cancers, ovarian cancer is responsible for more deaths in women than other reproductive system malignancies. It is also one of the most feared cancers, because it often is not detected until the disease has spread elsewhere in the body (metastasized). At that point, it becomes difficult to treat. However, there are ways that you can reduce your risk if you have a strong genetic or family history. Data suggests that ovarian cancer may start in the fallopian tubes, and risk-reducing surgery to remove the tubes and ovaries can be offered for women who are done with child-bearing. Also, you might catch ovarian cancer while it is still treatable if you see the doctor for subtle warning signs, including abdominal pain or bloating, pelvic pain, difficulty eating or feeling full quickly, and urgent or frequent urination. Because these symptoms are common, though, very few people who have them will have ovarian cancer.

Risk factors

The following are linked to an increased risk.

Age. The median age of diagnosis is 63.

Heredity. A family or personal history of ovarian or breast cancer indicates an increased risk. Women with a BRCA1 mutation have a 35% to 70% higher lifetime risk than women without the mutation; for BRCA2, the increase in risk is 10% to 30% by age 70. Another genetic syndrome, hereditary nonpolyposis colorectal cancer (see “Family history,” page 33), has also been associated with endometrial and ovarian cancer.

Postmenopausal estrogen use. Five or more years of estrogen as postmenopausal hormone therapy, without concurrent use of a progestin, has been shown to increase risk in several large studies.

Overweight or obesity. A large observational study indicated that a body mass index (BMI) of

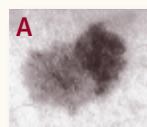
30 or greater increases the risk of ovarian cancer by 80% in postmenopausal women who have never taken hormones. You can find your BMI by entering your height and weight into the online calculator at www.health.harvard.edu/BMI.

Melanoma

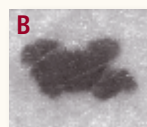
While melanoma makes up only 1% of skin cancer diagnoses, it's responsible for the majority of skin cancer deaths. It generally develops on parts of the body that receive a lot of sunshine, but it can also occur in areas that don't, such as the soles of the feet or the genital area. Without treatment, melanoma can spread to the lymph nodes and internal organs. It's a good idea to memorize the signs of moles that may have developed into melanoma (see “The ABCDEs of melanoma”).

The ABCDEs of melanoma

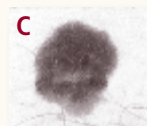
If a spot on your skin meets any of the criteria below, see a dermatologist right away.



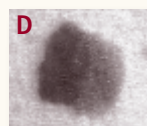
Asymmetry. Most moles have a round, symmetrical shape, but melanoma is asymmetrical, meaning one side may be different in shape than the other.



Border irregularity. Normal moles are round and typically have a clear border. In melanoma, the borders are shaggy, uneven, or indistinct, sometimes blending into the surrounding skin. A mole resembling the shape of cauliflower, for instance, should be checked out.



Color. Melanomas are usually very dark, but often are a mix of hues rather than one color.



Diameter. Most common moles are small, less than 5 millimeters, which is about a quarter of an inch. Melanoma grows, so talk to your doctor if the spot is larger than that.

Evolving. Look for any change in size, shape, color, or elevation, or any new symptoms such as bleeding, itching, or crusting.

Treatment involves removing the melanoma and a margin of healthy tissue. At the same time, the surgeon might do a sentinel node biopsy—removing the lymph node nearest the tumor to see if it contains any cancer cells. If it does, additional lymph nodes will be removed. In addition to surgery, standard treatments for melanoma include chemotherapy, radiation, and immunotherapy, which strengthens the immune system against the cancer. Newer targeted treatments, sometimes referred to as molecular targeted therapies or personalized medicine, promise to extend survival time for some people with metastatic melanoma. Several melanoma vaccines—designed both to prevent melanoma and treat existing cases—are being tested in clinical trials.

If melanoma is detected before it has spread to the nearby lymph nodes, the five-year survival rate is 99%. As with most other forms of cancer, if the tumor has spread to distant organs, survival is lower.

Risk factors for melanoma

A variety of factors put you at increased risk.

A history of sunburns or use of tanning beds. Damage from ultraviolet light is the main cause of melanoma. Just one session in an indoor tanning bed is dangerous; it can raise your risk of melanoma by as much as 20%.

Family history. Around 10% of people who develop melanoma have a first-degree relative—such as a parent, brother or sister, or child—who’s had the disease.

Fair skin, freckling, and light hair and eyes. If you have light skin, especially if you burn or freckle easily, you are at higher risk than someone with dark skin, since darker skin has higher levels of the protec-

How to reduce your risk of melanoma

- Stay out of the sun, especially when ultraviolet (UV) radiation is strongest (generally 10 a.m. to 2 p.m.).
- Wear sun-protective clothing, including a broad-brimmed hat.
- Use sunscreen labeled as having an SPF of at least 30 that blocks both UVA and UVB radiation.
- Avoid tanning beds.

tive pigment melanin. People with red or blond hair and blue or green eyes (who tend to have fair skin) are also at higher risk.

Moles, especially unusual ones. Although most moles will never cause problems, some do develop into melanomas. Watch out for new moles that appear after age 30 and for changes in existing moles. You are at particular risk if you have 20 or more moles larger than 2 millimeters (mm) across, five or more moles 5 mm across, or one or more irregularly shaped moles.

A personal history of skin cancer. If you’ve had skin cancer in the past—whether it was melanoma or one of the less deadly cancers known as basal and squamous cell carcinomas—you are more susceptible.

A weakened immune system. If your immune system is weak for any reason—such as a disease like HIV or an immunosuppressant medication you’re taking—it may not be as able to fight off any cancer, including melanoma.

Your age. Like most cancers, melanoma becomes more common as you get older.

Respiratory disease

Chronic obstructive pulmonary disease

Chronic bronchitis and emphysema—two lung diseases grouped together under the umbrella term chronic obstructive pulmonary disease (COPD)—develop gradually over many years, as the airways

become narrowed and the lungs lose their ability to exchange gasses with the blood efficiently when you breathe (see Figure 5, page 36). In most people with COPD, these problems stem from inflammation that occurs when something—most often cigarette smoke—repeatedly irritates the respiratory tract. The irritant can damage the cells that line the

airways and can cause changes in the glands and cells that normally produce small amounts of mucus to lubricate the airway walls. In response, the body unleashes a flood of inflammatory cells, which start a chemical cascade that further damages the airways and degrades lung tissue. The cells infiltrate the walls of the airways and trigger the production of mucus inside the airways, leaving less room for air to pass. They also prompt the release of enzymes that eventually break down the lung tissue. By the time you notice symptoms—namely, coughing and breathlessness—50% to 70% of lung tissue may be lost.

Risk factors

The following factors make a person more likely to develop COPD.

Smoking. Smoking—past and present—is the leading cause of COPD. Nearly 40% of people with COPD are current smokers, and 80% of COPD-related deaths are linked to smoking. Even secondhand smoke can put you at risk. An intriguing question is why more smokers don't get COPD. Only 15.2% of smokers and 7.6% of former smokers are reported to have COPD, strongly suggesting that genetic factors play a role as well.

How to reduce your risk of COPD

- Shun cigarettes (see “Kick the smoking habit,” page 45).
- Reduce your exposure to environmental toxins and fumes.

Other irritants. Many years of exposure to irritants such as air pollution, toxic fumes, industrial smoke, and dust also increase risk of COPD.

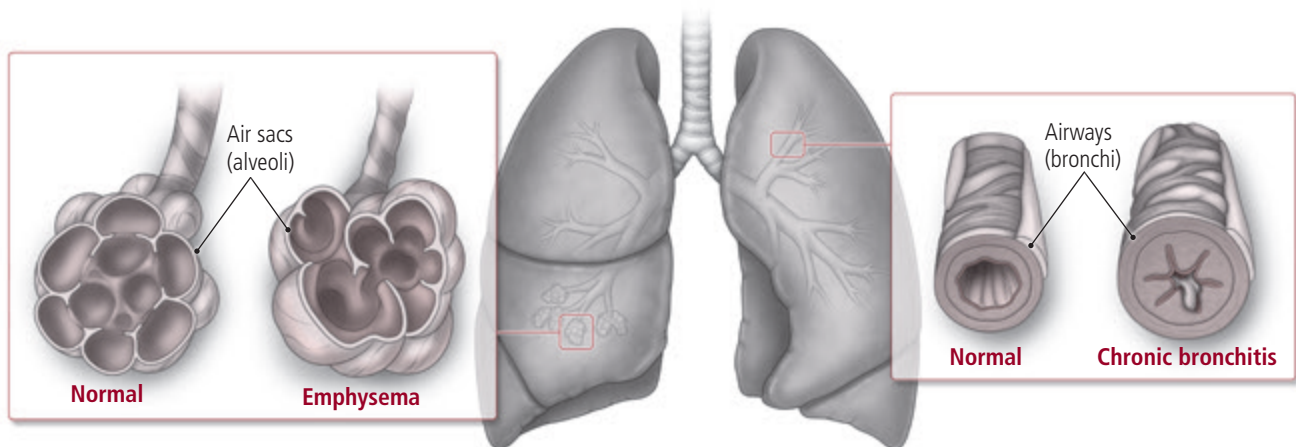
Genetics. Alpha-1-antitrypsin (A1AT) deficiency is an inherited condition involving a defect in the gene that controls the production of A1AT, a protein that neutralizes an enzyme that degrades lung tissue. When the gene is defective, the level of A1AT is about 15% of normal, giving proteases free rein to wreak havoc on the lungs. In all, A1AT deficiency causes 3% of cases of emphysema. Smokers who have the genetic defect are almost certain to get emphysema.

Influenza, pneumonia, and COVID-19

Influenza (flu) is caused by a highly infectious respiratory virus and is usually spread from one person to the next when the infected person coughs or

Figure 5: Lung damage in emphysema and chronic bronchitis

COPD encompasses two distinct diseases—emphysema and chronic bronchitis—both of which make breathing more difficult, albeit in different ways. Most people with COPD have components of both.



In emphysema, walls of the air sacs (alveoli) break down, leaving fewer, bigger air sacs with less surface area to allow exchange of oxygen and carbon dioxide between the lungs and the blood.

In chronic bronchitis, the large airways, called bronchi, become inflamed, swollen, and clogged with mucus. The same types of damage occur in the small airways, called bronchioles (not shown here).

Flu and pneumonia vaccine recommendations

Vaccines significantly reduce the risk of pneumonia and influenza.

A pneumococcal polysaccharide vaccine called pneumococcal 13-valent conjugate (abbreviated PCV13, or Pevnar 13) offers protection from pneumococcal pneumonia, blood infection, and meningitis. Recommended at age 65, it is to be followed by a second vaccine—pneumococcal 23-valent polysaccharide (PPSV23, or Pneumovax 23)—a year or more later. Both vaccines are also advised for younger people who have chronic organ disease or compromised immunity.

The seasonal flu shot, containing killed flu virus, is recommended for all adults. A higher dose of the flu vaccine is advised for people over 65.



sneezes. Droplets containing the flu virus can travel as much as six feet, so you don't need to be standing right next to someone to become infected. Less often, you can become infected after touching a contaminated surface such as a doorknob.

In healthy people, flu can hit with a wallop, causing fever, chills, fatigue, muscle aches, and maybe a headache or sore throat. In older adults, pregnant women, or people with other medical conditions, flu can be deadly. One of the challenges in fighting flu is that viruses mutate so rapidly that older vaccines don't necessarily work against them. That's why the flu shot has to be reformulated every year.

Pneumonia also develops because of a respiratory infection, although the cause can be a virus, a bacterium, or even an inhaled chemical. There are two main types of pneumonia. Viral pneumonia is caused by respiratory viruses such as the flu virus or the virus that causes COVID-19.

Bacterial pneumonia can be triggered directly by any one of dozens of bacteria, or it may emerge as a complication of flu or another viral infection. However it develops, pneumonia causes body aches, fever, chills, cough, difficulty breathing, and sometimes death.

COVID-19

Like the flu, COVID-19 is a respiratory virus that rides along on tiny droplets and aerosols breathed, coughed, or sneezed into the air by an infected person. Typical symptoms include:

- cough, sore throat, congestion, or runny nose
- fatigue
- headache or body aches
- fever
- appetite loss, vomiting, and diarrhea
- changes in the senses of smell and taste.

Most COVID infections are mild to moderate and can be weathered at home. However, severe cases of COVID can lead to hospitalization and death. Most COVID deaths have occurred in people 65 and older. Also at higher risk are people with underlying health conditions, such as weakened immune systems, diabetes, heart disease, lung disease, and overweight or obesity, or who have limited access to health care or other resources.

If you contract the virus and are at high risk of complications, ask your doctor if you should take Paxlovid (nirmatrelvir and ritonavir). It can dramatically cut the risk of serious complications such as hospitalization when taken within five days of symptom onset.

Some people who contract the virus may continue to have symptoms or develop new ones months after the initial infection. Long-term symptoms can include deep exhaustion, chest pains, muscle aches,



Staying up to date on vaccines significantly lowers the risk of developing serious complications from respiratory infections including the flu, pneumonia, and COVID-19.

► How to reduce your risk of flu, pneumonia, and COVID-19

- Get vaccinated (see Table 6, page 49).
- Wash your hands or use hand sanitizer frequently throughout the day.
- Consider wearing a mask in crowded indoor settings, especially during the fall and winter or when rates of respiratory viruses spike.
- If you get the flu or COVID-19, see your doctor so you can get prompt treatment and avoid having it turn into pneumonia or other serious complications.
- Shun cigarettes (see “Kick the smoking habit,” page 45).

and memory problems, among others. This syndrome has come to be known as long COVID or post-COVID conditions.

Getting an updated COVID vaccine is crucial. (Check www.cdc.gov for the latest guidance.) The CDC notes that wearing a mask may be a wise strategy for some people and can be particularly helpful when rates of respiratory viruses, including COVID, are high in your area. Cleaning your hands often and avoiding sick people as much as possible is always advisable.

Risk factors

Being aware of the following risk factors can help you avoid catching flu or pneumonia. Many of these factors also appear to influence your risk of either getting COVID-19 or developing a severe case of it.

Smoking. Smokers are more likely to become seriously ill with the flu than nonsmokers, as are people exposed to secondhand smoke. Smoking also makes it more likely that a person with COVID-19 will develop a serious case.

Crowded and close environments. Spending time in places like bars, restaurants, nursing homes, gyms, and theaters increases the risk of contracting the flu and COVID-19.

Age. Children ages 2 and under face the highest risk of flu, followed by people 65 and older. People 65 and older remain at higher risk for serious complications.

Existing medical conditions. Cancer, neuromuscular diseases, chronic lung conditions, and cardiovas-

cular diseases, as well as immunosuppressive disorders and drug therapy that can suppress the immune system: all increase the risk of getting the flu. Other conditions, including diabetes and obesity, also increase your risk of having a serious case of COVID-19.

Dental plaque. Even the bacteria that inhabit dental plaque can migrate to the lungs to cause pneumonia. The risk is greatest in people on ventilators and in nursing homes.

Alzheimer's disease

Alzheimer's disease exacts a disproportionately heavy toll on women. Almost two-thirds of the nearly seven million Americans with Alzheimer's are women. And women make up nearly two-thirds of those caring for someone with Alzheimer's.

It's not clear what sequence of events in the brain causes Alzheimer's to develop. According to the leading theory, small fragments of soluble beta-amyloid protein in the brain may be the toxic factor, as well as another protein called tau; these trigger a series of biochemical changes leading to the death of brain cells. Two telltale signs of the disease are plaques made of beta-amyloid and neurofibrillary tangles made of tau—abnormalities that can now be detected with specialized PET scans of the brain.

Whatever the cause of Alzheimer's, the results are clear. The brain gradually loses cells (neurons) and the connections (synapses) between neurons that enable memory and other mental functions. Levels of

► How to reduce your risk of Alzheimer's

- Keep your blood pressure within healthy limits (see “High blood pressure (hypertension),” page 27).
- Keep your cholesterol in check (see “Unfavorable cholesterol profile,” page 27).
- Watch your weight (see “Measure your waist,” page 46).
- Try to avoid developing diabetes (see page 39).
- Sidestep a sedentary lifestyle (see “Keep moving,” page 42).
- Wear a helmet while riding a bike.

brain chemicals known as neurotransmitters, which carry complex messages among billions of neurons, also decrease. In advanced Alzheimer's disease, the dramatic loss of neurons causes the brain to shrink.

After the symptoms first appear, people live anywhere from two to 20 years in an increasingly dependent state that exacts a staggering emotional, physical, and economic toll on families.

There is no cure for Alzheimer's, but early diagnosis can still make a difference. Drugs are available that may temporarily stabilize or slow the progression of cognitive symptoms. These medications work best in the early stages of the disease.

Risk factors

Some of the risk factors for Alzheimer's, such as age and genetics, are things you cannot change. But many of the risk factors for the condition are things you can modify, including smoking, midlife obesity, and physical inactivity.

Age. Risk usually rises after age 65 and doubles every five years thereafter.

Genetics. Heredity is a decisive factor in only a small number of families, primarily those with early-onset Alzheimer's, in which the disease strikes people in their 40s and 50s. Mutations in three genes are known to cause this type of Alzheimer's: the amyloid precursor protein gene, presenilin 1, and presenilin 2. All three genetic mutations increase the production of beta-amyloid, which is deposited in the plaques found in Alzheimer's disease.

In late-onset Alzheimer's, the more common form of the disease, the genetic link is not as strong. But APOE4—one of several alleles (versions) of the gene that encodes for apolipoprotein E, a key protein in some biological processes—influences susceptibility to Alzheimer's. APOE4 has been found in about 40% of people with late-onset Alzheimer's, and a test for this genetic marker is widely available. However, the presence of APOE4 doesn't necessarily indicate that you will develop the disease, so it is not recommended for healthy people without dementia to use in assessing their Alzheimer's risk.

Cardiovascular risk factors. Many studies show that the same things that harm the heart and blood

vessels also increase the risk of developing Alzheimer's disease. These include high blood pressure, high cholesterol, obesity, and smoking. All of these factors also increase the risk of having repeated small strokes, which can be a direct cause of vascular dementia, the second most common form of dementia after Alzheimer's.

Diabetes. People with diabetes are twice as likely to develop Alzheimer's disease as those without diabetes. Research indicates that this increase in risk may reflect a shared mechanism—a resistance of cells to the effects of insulin, the hormone that enables cells in the body to use blood sugar.

Head injury. The microscopic changes in the brains of boxers and football players with dementia resemble those in Alzheimer's disease—especially the presence of neurofibrillary tangles. The observation has led researchers to wonder if brain injury might be a factor in the disease.

Sedentary lifestyle. An analysis of 16 prospective studies on the association between physical activity and dementia found that participants who were most active had a 45% lower risk of Alzheimer's compared with those who were least active.

Diabetes

Blood sugar (glucose) is the fuel that the body uses for energy. Diabetes develops when the body's ability to turn glucose into energy breaks down and blood glucose levels rise to dangerously high levels. Left untreated or poorly managed, it can lead to serious long-term complications, including kidney failure, amputation of lower limbs, and blindness.

There are two types of diabetes. About 5% to 10% of people with diabetes are diagnosed with type 1 dia-

► How to reduce your risk of type 2 diabetes

- Watch your weight (see "Measure your waist," page 46).
- Boost your activity level. Regular exercise protects you, even if you don't lose weight (see "Keep moving," page 42).
- Improve your diet (see "Adopt a pattern of healthy eating," page 43).

betes, which develops after the body's immune system mistakenly destroys cells in the pancreas, so that this organ can no longer produce the insulin necessary to metabolize blood sugar. This form of the disease usually develops before age 20.

Far more common—and often preventable—is type 2 diabetes. It tends to occur in adults over 40, and its incidence increases with age, although recent obesity trends have put younger people at greater risk as well.

Type 2 diabetes takes years to develop. The process begins when cells of the body become less responsive to insulin. To keep blood sugar levels under control, the pancreas compensates by pumping out more insulin. At first, blood sugar levels stay within a normal range. But for some, the insulin-producing cells eventually fail to keep up with the increased demand. As a result, blood sugar levels rise, resulting first in a problem known as prediabetes, then finally progressing to full-blown diabetes.

A doctor generally diagnoses the disease when fasting blood sugar is 126 mg/dL or higher on two separate tests; when an oral glucose tolerance test shows a blood sugar reading of 200 mg/dL or more two hours after drinking a sugary liquid; or when HbA1c, a measure of average blood sugar over the previous two to three months, is 6.5% or higher. You may also have symptoms of high blood sugar, such as blurry vision, excessive thirst, and frequent urination.

Risk factors

Reducing controllable risk factors can often prevent type 2 diabetes from developing.

Age. Predominantly a disease of later life, type 2 diabetes generally develops after age 40. People over age 65 are at particularly high risk.

Family history. Having a first-degree relative with diabetes raises the risk. If one of your parents was diagnosed with type 2 diabetes before age 50, your risk of getting the disease is roughly one in seven. If your parent was diagnosed after age 50, the risk is about one in 13. If both of your parents developed type 2 diabetes, your odds are about one in two.

Ethnicity. The disease is far more common among African Americans, Hispanics, Asian Ameri-

cans, Pacific Islanders, and Native Americans than among whites. At least part of the reason involves challenges in accessing healthy food and equitable health care.

Obesity or overweight. Of the more than one million Americans who develop diabetes annually, about 80% are overweight or obese. The distribution of body fat seems to be important. People who tend to store fat around the waist are more likely to develop diabetes than those who put it on around the hips.

Sedentary lifestyle. People who don't get regular exercise have an increased risk for diabetes.

High blood pressure. If you have high blood pressure, you have double the risk of being diagnosed with diabetes. It's not that high blood pressure causes diabetes, but that the two often occur in the same population—that is, sedentary people who are overweight. Both conditions also raise your risk of developing and dying from heart-related problems.

Polycystic ovary syndrome. Women with this hormonal disorder usually have insulin resistance or other diabetes risk factors.

Gestational diabetes. This form of the disease occurs during pregnancy and usually resolves after delivery, but many women who had this problem while pregnant later develop full-blown diabetes (usually five to 10 years after the pregnancy).

Prediabetes. This is defined as having fasting blood glucose levels of 100 to 125 mg/dL or oral glucose tolerance test levels of 140 to 199 mg/dL. Around 15% to 30% of people with prediabetes develop diabetes within five years. But you can still prevent diabetes, primarily by developing a healthier lifestyle. The Diabetes Prevention Program study has demonstrated that it's possible to dramatically reduce the progression of prediabetes to full-fledged diabetes simply by losing 5% to 7% of your body weight and increasing your physical activity. In the initial study, the risk reduction was an impressive 58%. In the 15-year follow-up, as people aged, it was 28%. Participants at high genetic risk for diabetes benefited just as much as others.

Medication use. Some drugs can increase insulin resistance or decrease insulin secretion. Either way, the result is higher levels of glucose in the blood-

stream. These medications include diuretics, corticosteroids, beta blockers, and a class of drugs called atypical or second-generation antipsychotics, originally developed to treat schizophrenia.

Accidents

Injuries are a leading cause of death for women as well as men, according to the CDC. Accidents that cause many of these injuries include falls, motor vehicle accidents, and poisonings.

Falls. Falls are a leading cause of injury-related deaths in people over age 65. The risk of falling, and that of incurring a head injury or fracture in a fall, both increase with age.

Traffic accidents. Thousands of women (and men) die in motor vehicle accidents each year.

Substance misuse and abuse. Both prescription and street drugs (in particular, opioids) have the potential to lead to fatal consequences. The CDC lists three main drug-related causes of death in adults:

How to reduce your risk of accidents

- Don't drink and drive.
- Buckle up when you drive, and take commonsense precautions, such as driving within the speed limit and not texting while you drive.
- Tell your doctor about all the medications you are taking so you can make substitutions for those most likely to affect your balance and avoid accidental drug interactions or overdoses.
- At home, remove tripping hazards such as loose rugs, install handrails or grab bars if necessary, and ensure that you have proper lighting so you can see obstacles clearly.

overdoses of drugs taken for non-medical reasons; poisonings from prescription drugs taken in error or at the wrong dose; and unanticipated effects of drugs (such as drug interactions). Alcohol poisoning can also be fatal, and alcohol in general increases the risk of accidents and injuries. ♥

Domestic violence

Domestic violence—also known as intimate partner violence, or IPV—is the leading cause of violence against women of all races and social classes, affecting as many as one in three women at some point. In fact, half of all women who are murdered in the United States are killed by a current or former male partner.

If you are experiencing IPV, resources are available to help you. Don't be afraid to take action.

Formulate a safety plan. It can be difficult to know what to do in a terrifying situation. Having a plan gives you a road map to follow in an emergency, when you may be too frightened to think clearly. The National Domestic Violence Hotline offers tips on creating a plan on its website at <https://www.thehotline.org/plan-for-safety>. The extensive guidance there will also help you stay safe

while planning to leave an abusive partner and in the aftermath of your departure. It will help you minimize the risk to your children and pets as well.

Seek medical treatment. Don't be afraid to go to the emergency room if you are injured. Medical staff are trained to deal with these sensitive matters.

Find support. There are a number of organizations that can provide you with support. Call the National Domestic Violence Hotline toll-free at 800-799-SAFE (7233) or 800-787-3224 for TTY. If you are unable to speak safely, you can log on to www.thehotline.org or text START to 88788. These resources are available 24/7, and the agency's staff can work with you to find resources in your area.

10 steps to a longer and healthier life

Healthy lifestyle changes are a little like one-stop shopping, in that a single change has the potential to yield numerous benefits. For example, not only does physical activity lower your risk of heart disease and diabetes, but it's also linked to having fewer colds. Improving your sleep lessens your risk of diabetes and helps ward off weight gain. So stop making excuses and start making the changes that matter most for good health.

1. Keep moving

Lack of physical activity is an independent risk factor for nearly all of the diseases that are most likely to kill or disable you—heart disease, stroke, multiple types of cancer, diabetes, osteoporosis, depression, Alzheimer's disease. In the long-running Framingham Heart Study, women who were very active enjoyed three-and-a-half more years of life than their sedentary counterparts.

The Physical Activity Guidelines for Americans from the U.S. Department of Health and Human Services reflect the latest evidence-

based advice on the amount and type of activity most people should aim for:

- **Aerobic activity.** Aerobic exercise includes any activity that gets your heart and lungs working harder, such as walking, jogging, or swimming. The federal guidelines recommend that all adults aim for a minimum of 150 minutes of moderate-intensity aerobic activity per week, or 75 minutes of vigorous-intensity exercise. All types of activity count, including dancing, gardening, and taking the stairs instead of the elevator.
- **Muscle strengthening.** Set aside two or more days a week, with at least 48 hours of rest between, to do strength exercises at moderate or greater intensity for all the major muscle groups (legs, hips, back, chest, abdomen, shoulders, and arms). You can do this by lifting weights or using your own body as the "weight" (for example, with push-ups).
- **Balance exercises.** Older women at risk for falls may also want to do balance exercises, such as



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Staying active can lower the risk of many serious diseases. It may even add years to your lifespan.

standing on one foot for increasing periods of time or practicing tai chi.

In general, the guidelines endorse the maxim that some exercise is better than none—but more is even better. Specifically:

Move more, sit less. Greater time spent sitting is associated with poorer health. Try to take regular breaks during the day to stand up and move around.

The more you move, the better. Getting 150 minutes a week of moderate aerobic activity is, for many people, an achievable goal. Doubling the total to 300 minutes of moderate aerobic activity boosts the benefits, as does exercising at a higher intensity (provided you know your limits and don't overdo it). There is no age limit on this advice; just do as much as you can and work up gradually.

If you don't have time for long workouts, break it up: take 10 minutes several times a day to walk around the block or do some yoga.

2. Adopt a pattern of healthy eating

"Superfoods" get a lot of attention, but what really counts is having an overall healthy dietary pattern that you adhere to day in, day out. Scores of observational studies have identified the dietary patterns that are associated with lower rates of heart disease, cancers, and dementia. To encapsulate this information in a simple graphic, nutrition scientists and epidemiologists at

the Harvard T.H. Chan School of Public Health and Harvard Medical School developed the Harvard Healthy Eating Plate (see Figure 6).

Here are the main points:

Half your plate should contain fruits and vegetables. You can't go wrong if you aim for color and variety. The greater the variety, the greater the range of nutrients you take in. And the pigments do more than provide eye appeal: many are antioxidants that reduce inflammation. (Note that for these purposes, potatoes—which are quick to raise blood sugar levels—don't count as vegetables.)

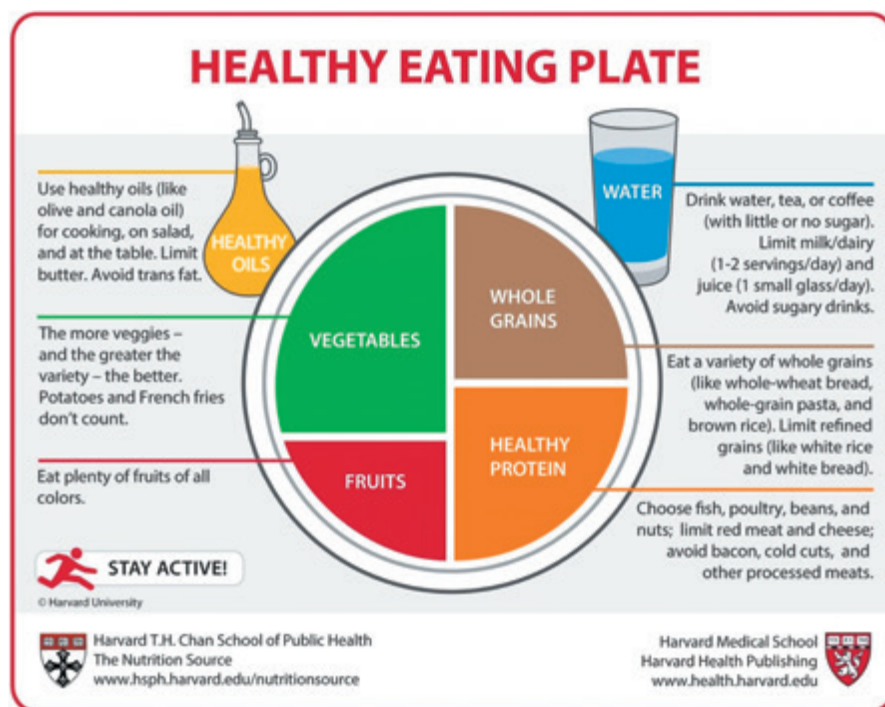
A quarter of the plate should be filled with whole grains. Whole wheat, barley, oats, and brown rice are more nutritious and



Load up on colorful produce to boost your intake of health-promoting antioxidants.

contain more fiber than refined grains. They have less effect on blood sugar and insulin than white bread, white rice, and other refined grains. Although whole grains may

Figure 6: The Harvard Healthy Eating Plate



take a little longer to cook, they offer more interesting flavors and textures than their refined counterparts. If you're feeling adventurous, try "ancient grains" like amaranth, quinoa, and teff, which have been unchanged for centuries.

The final quarter should consist of healthful sources of protein. Fish, chicken, beans, and nuts are all healthy, versatile protein sources. Limit red meat, which has been linked to an increased risk of cardiovascular disease and cancer, to one or two servings a week. Avoid processed meats, such as bacon and sausage, which have been designated carcinogens by the World Health Organization.

Include healthy plant oils. Choose vegetable oils like olive, canola, sunflower, and peanut in cooking. Avoid partially hydrogenated oils, which increase the risk of cardiovascular disease; they are commonly found in nondairy creamers, packaged baked goods, snack foods, vegetable shortenings, and stick margarines. And remember that low-fat does not always mean healthy; low-fat dressings and sauces often contain more salt and sugar than higher-fat versions.

Drink water, coffee, or tea. Skip sugary drinks, limit milk and dairy products to one to two servings per day, and limit fruit juice to a small glass per day.

3. Don't skimp on sleep

The average adult needs seven to nine hours of sleep daily for opti-

mum health and function. However, more than 60% of women regularly fall short of that goal. In some cases, lack of sleep results from insomnia or other underlying conditions that may require medical attention. But most women burn the candle at both ends—consistently failing to get to bed early enough or stay there long enough. Women often find it difficult to make up sleep lost while juggling family and career. Menopause can add its own challenges.

Many studies show that sleep shortfalls can lead to a range of health problems, from the mild (being more likely to catch a cold) to the more consequential (having a higher risk of high blood pressure, heart disease, obesity, and diabetes). Think of sufficient sleep as a pillar of health—just as important as eating a healthy diet and getting regular exercise.

Here are some ways to get more shut-eye:

Make sleep a priority. In this 24/7 world, it's tempting to prioritize work and other activities over sleep. Experts at the National Sleep Foundation recommend scheduling sleep like any other essential activity on your to-do list. "Don't make it the thing you do only after everything else is done," they advise.

Create a sleep sanctuary. Reserve your bedroom for sleep and intimacy. Keep it on the cool side. Banish the television, computer, cellphone or tablet, and other diversions from the space.

Stick to a schedule. Try to go to bed and get up at the same time every day—at the very least, on weekdays. This helps keep your sleep/wake cycle synchronized. If you need to make up for lost sleep, try taking a nap in the early afternoon. A nap of 15 to 20 minutes is best, but it should definitely not exceed 40 minutes.

Get regular exercise. It's a proven way to increase the amount of time you spend in deep sleep, the type that is most restorative. Exercising outdoors in the morning is ideal, because bright natural daylight helps reset your circadian rhythms. Try to avoid exercising within two hours of bedtime, because exercise acts as a short-term stimulant and may make it harder for you to drift off.

Cut off caffeine early. If you have problems sleeping, avoid caffeine from any source—including chocolate—after 2 p.m., or noon if you are especially caffeine-sensitive. Caffeine can stay in your body for up to 12 hours.

Nix the nightcap. Alcohol may make you fall asleep more quickly, but the soporific effects disappear after a few hours and drinkers have frequent awakenings during the night.

If you're able to get enough sleep but don't feel refreshed in the morning, talk to your doctor. Many common medical conditions, from depression to obstructive sleep apnea (brief cessations in breathing during sleep), could be responsible.

4. Kick the smoking habit

There are few things you can do that will have such immediate and lasting benefits as giving up smoking. Within 12 hours after you stop smoking, your blood transports oxygen more efficiently. By the end of the first week, your circulation improves. By the one-month mark, your lungs are functioning more efficiently. Over the years, your risk for lung cancer, stroke, and heart disease will drop by at least half. And in 15 years, you'll have erased your excess risk for heart disease.

Quitting is rarely easy, but these tips may help:

Set a quit date. You might plan to quit on a special day, but avoid making it during a high-stress period.

List your motivations. Write down your reasons for wanting to quit and refer to it whenever you're feeling tempted.

Avoid triggers. Any addiction is learned, in that the brain comes to associate sights, smells, and situations with the addictive behavior or drug. Identify triggers that make you want to smoke and avoid them until you feel stronger.

Find a new habit. Find a habit to replace smoking. Go for a walk after dinner, or do something else to distract yourself.

Deal with withdrawal. Nicotine replacement therapy comes in many forms: patches, gum, nasal sprays, inhalers, and lozenges. If this approach doesn't work, ask your doctor about medication.

Keep on trying. Your first attempt might not be successful, but don't give up. It often takes a few attempts to successfully quit.

5. Rethink your drink

Recent studies have debunked the longstanding notion that moderate drinking has health perks. Several expert groups, including the American Cancer Society, now say that avoiding alcohol entirely is the best option. If you do choose to drink, limit it to no more than one drink per day (see Figure 7).

Too much alcohol at any age can raise the risk of high blood pressure, stroke, and car accidents, as well as pancreatic and liver disease. Alcohol also raises the risk of numerous types of cancer, including those of the breast, colon, esophagus, larynx, pharynx, oral cavity, and liver.

As you age, drinking becomes

even riskier. Older adults have increased sensitivity to the effects of alcohol because they typically metabolize alcohol more slowly. Lean body mass also declines with age, and with less muscle to absorb alcohol, older adults feel the effects of alcohol more quickly. Combined with other physical changes in the body due to age, older adults who drink alcohol are susceptible to falls, bone fractures, and other unintentional injuries. Given that many older adults are taking multiple prescription medications, the dangerous and sometimes fatal consequence of mixing medications with alcohol are another important concern for this age group.

Women are especially vulnerable. Compared to men, women produce less of an enzyme that breaks down alcohol in the stomach before it's absorbed into the

Figure 7: What is a standard drink?

- 1½ ounces (a jigger) of 80-proof liquor (bourbon, brandy, gin, rum, scotch, tequila, vodka, or whiskey)
- 2 to 3 ounces of fruit, coffee, chocolate, or other flavored liqueurs (cordials)
- 3 ounces of fortified wine (sherry, port, Marsala, or Madeira)
- 4 to 5 ounces of table wine
- 12 ounces of regular or light beer



blood. Moreover, women’s bodies have a lower proportion of water (which dilutes alcohol) to fat (which retains it). Alcohol therefore remains in the female system at higher concentrations for longer periods of time and has more contact with the brain and other organs. Compared to men, women become intoxicated faster, develop addiction earlier, and suffer health damage sooner.

6. Measure your waist

Excess body weight increases your risk for more than 195 different health problems. These conditions include some of the leading causes of death among women—heart disease, stroke, breast cancer, and diabetes—as well as less serious ailments such as arthritic knees and gallstones.

Body mass index (BMI) has long been the go-to tool for guiding weight management. However, experts have come to realize this

easy-to-use calculation may not result in a true reflection of the health risks associated with excess body fat, since BMI does not distinguish between body fat that accumulates directly under the skin versus visceral fat, which collects around organs in the abdomen (see Figure 8). The problem with visceral fat is that it’s metabolically active, producing substances that spur inflammation and increase insulin resistance. Thus an “apple” body shape—particularly having a waist measurement of 35 inches or more—puts a woman at higher risk for heart disease and diabetes.

For this reason, researchers are increasingly looking at waist-

How to measure waist-to-hip ratio

To calculate your WHR, first use a measuring tape to check the circumference of your waist at the level of the belly button. Then measure your hips at the widest part. Divide your waist measurement by your hip measurement to assess your risk, or you can use this online tool: www.health.harvard.edu/WHratio.

WHR and health risk		
MALE	FEMALE	RISK
0.95 or below	0.80 or below	low Risk
0.96 to 1.0	0.81 to 0.85	moderate Risk
1.0+	0.85+	high Risk

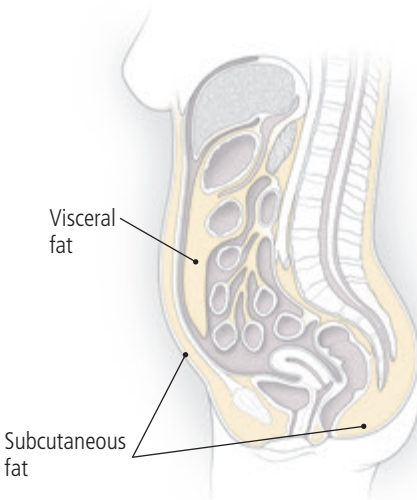
to-hip ratio (WHR) as a more accurate measurement for assessing metabolic risk. A 2023 study published in *Diabetes and Endocrinology* that followed over 380,000 adults in the United Kingdom found WHR to be a more accurate and consistent indicator of all-cause mortality risk than BMI.

Where a woman puts on fat is influenced by heredity and hormones. At menopause, estrogen production decreases. At the same time, the ratio of androgens (male hormones present in small amounts in women) to estrogen increases—a shift that’s been linked in some studies to increased abdominal fat after menopause.

If your doctor recommends that you lose weight, focus on improving diet quality, specifically on avoiding processed foods, opting instead for foods in their natural form. In addition, follow the exercise and dietary guidelines

Continued on page 48

Figure 8: Types of body fat



The excess pounds that tend to creep on at midlife often settle around the belly and hips. Fat that accumulates just under the skin, called subcutaneous fat, is found in the hips, thighs, and buttocks, as well as between the abdominal wall and skin covering your belly. Yet more of the fat in the abdominal area is visceral fat. Located around the abdominal organs and inside the abdominal cavity, visceral fat appears to increase the risk of insulin resistance, which can set the stage for type 2 diabetes.

Table 5: Routine screening tests for women ages 50 or older

Screening tests check for diseases and conditions that may not yet be causing symptoms. If you already have a particular disease or condition—or are at high risk for one—more frequent testing may be appropriate.

TEST	RECOMMENDATION
Blood pressure testing	You should get checked yearly if you are over age 40. Your doctor may recommend more frequent screenings if you have certain conditions or risk factors.
BRCA risk testing	If you have family members with breast, ovarian, tubal, or peritoneal cancer, you should consider having a blood test to detect mutations in breast cancer susceptibility genes BRCA1 and BRCA2.
Breast cancer screening	Some disagreement exists over the frequency of screening (see Table 4, page 31). Currently, the U.S. Preventive Services Task Force recommends women ages 50 through 74 get a mammogram every two years, and that women 75 or older discuss their screening schedule with their doctors. The American Cancer Society recommends the option to start annual screening between ages 40 and 45, and continue with annual screening from age 50 through age 54. Women age 55 and older may continue with annual screening, or change to a biannual screening schedule.
Cholesterol testing	You should begin regular cholesterol testing at age 45 if you do not have known heart disease risk factors. Ask your doctor how often you should get repeat screenings. Frequency will depend on your lifestyle and any other health conditions you may have.
Colorectal cancer screening	Get screened for colorectal cancer starting at age 45. Talk to your doctor about which screening test—fecal occult blood, sDNA-FIT, sigmoidoscopy, virtual colonoscopy, or colonoscopy—is best for you and how often you need it.
Diabetes screening	Get screened for diabetes during your regular preventive health visits. The National Institutes of Health recommends more frequent screening if you are overweight or obese, have a family history of diabetes, or have cardiovascular risks such as high blood pressure, prediabetes, or a history of heart disease.
Hepatitis B virus testing	You may need a blood test for hepatitis B virus infection if you are at high risk for infection.
Hepatitis C virus testing	Adults ages 18 and older should be screened at least once. Pregnant women should be screened during each pregnancy. People in other age groups may also be screened if they are considered high risk.
HIV testing	Everyone between the ages of 15 and 65 should be screened at least once. Younger and older individuals should also be screened if they are considered high risk.
Lung cancer	If you are age 50 to 80, have smoked the equivalent of a pack of cigarettes a day for 20 years, and currently smoke or have quit within the past 15 years, you should have a low-dose CT scan annually.
Osteoporosis	Have a dual energy x-ray absorptiometry (DEXA) scan at least once after age 65. Talk to your doctor about repeat testing. Younger individuals should also get a DEXA scan if they are high risk according to a clinical assessment tool.
Pap and HPV tests	If you have a cervix, get a Pap test and human papillomavirus (HPV) test together every five years until age 65. Ask your doctor if you need to continue testing after age 65.
Sexually transmitted infections	Get tested for chlamydia, gonorrhea, and syphilis if you are at increased risk. Also see screening recommendations for hepatitis C and HIV.
Tests at well-woman and preventive care visits	Insurance plans are required to cover a yearly well-woman visit for women under 65. Medicare covers a preventive care visit for women 65 or older. During these visits, you should be screened for high blood pressure, obesity, alcohol abuse, and depression.

Source: U.S. Preventive Services Task Force.

Continued from page 46

outlined earlier in this section. Participants in the National Weight Control Registry—a database of more than 10,000 people, 80% of whom are women, who have lost 30 pounds or more and kept it off for at least a year—report exercising approximately one hour a day (primarily through brisk walking), while 98% have modified their diet in some way. They also keep close tabs on their weight, hitting the scale at least once a week, while 62% report watching less than 10 hours of television a week.



© Ariel Skelley | Getty Images

Volunteering can be a good way to stay active and remain engaged in your community. It can also help you strengthen social ties or forge new ones, which may help you ward off loneliness and solidify a sense of purpose later in life.

7. Take charge of your own health

To help ensure you're taking the best possible care of your own health, be persistent and follow these tips:

Form a partnership. At midlife and beyond, good health is increasingly dependent on successful collaboration with your doctors. Since you are likely to be making more medical visits, find a primary care clinician with whom you feel comfortable.

Get screened. Work with your doctor to devise a schedule for screening tests and clinical exams (see Table 5, page 47). Take prescribed medications as directed; don't stray from the designated dosage without talking to your doctor. Find out what you should expect from every new medication and procedure.

Get vaccinated. Vaccines are not just for kids (see Table

6, page 49). There are a number of vaccines that you should get in your adult years to protect against diseases like influenza and shingles. Check with your doctor to make sure you're up to date.

Manage your medical conditions. You may not notice any symptoms from conditions like high blood pressure, high cholesterol, diabetes, or glaucoma. But it's important to work with your doctor to keep these under control. Problems can develop many years before serious consequences become apparent.

Pay attention to your body. Keep an eye on new spots and bumps and note any changes in your regular patterns, be they in appetite, sleep, energy, bowel habits, or mood. It's easy to write off many changes as consequences of aging, but they could signal underlying disease.

8. Stay connected

Women are living longer, and with aging comes the risk of isolation, as you retire and as your friends, partners, and relatives go into decline or die. Meanwhile, friends and neighbors may move away. This means that, as you age, you need to work harder to maintain and bolster your social connections in order to enhance your health into old age. Informal socializing with friends is of particular importance.

Numerous studies have charted the positive influence of social networks on health later in life. For example, the Baltimore Longitudinal Study on Aging and the Georgia Centenarian Study have both found that older adults who remain socially active live longer and healthier lives than their solitary counterparts. Building connections in your local community

is important, since studies show that having trusted friends and neighbors makes it more comfortable for you to walk and stay active in the neighborhood, get help with chores, and share rides to keep up with health screenings.

Try the following tips to make new social connections:

Reach out. Set up a regular routine for socializing with particular friends. Getting together for a

movie or game night once a week sets up a structure for lasting personal relationships.

Find common ground. Connect with others through shared interests or religious affiliations.

Choose an exercise buddy. You're more likely to stick with an exercise program, and you'll also get the benefits of socializing. Taking a daily morning walk with a friend is an easy option.

Join a group or club. Do you like reading, crafting, or chess? Join a book group or some other group of people who share your interest. Check with universities, the YMCA, community centers, and your local library.

Take a class. Learn a new language, or try ballroom dancing or pottery making, and you'll meet new people in the process.

Volunteer. Donating your time to a good cause offers double benefits: you get to meet a whole new circle of contacts while doing something to help others.

Table 6: Immunizations for women ages 50 or older

DISEASE	RECOMMENDATION
Chickenpox	If you did not have chickenpox as a child, you should get two doses of varicella vaccine unless you have a weakened immune system.
COVID-19	The CDC recommends that all people over 5 years of age follow a regular COVID vaccination schedule. Check www.cdc.gov for the latest recommendations.
Flu	You should get a flu shot every year. A high-dose version is available for people over 65.
H flu (caused by <i>Haemophilus influenzae</i> type b, or Hib)	You should have the Hib vaccination if your spleen has been removed or if you have a weakened immune system. Your doctor may also recommend it if you have other risk factors or chronic conditions.
Hepatitis A	You should have this vaccination if you have chronic liver disease. Your doctor may also recommend it if you have other chronic conditions.
Hepatitis B	All adults through age 59 should have this vaccine. Your doctor may recommend it if you are age 60 or older, or if you have other risk factors or chronic conditions.
Meningitis (caused by <i>Neisseria meningitidis</i>)	You should have a MenACWY or MenB vaccination if your spleen has been removed. Your doctor may also recommend one of these vaccines if you have other risk factors or chronic conditions.
Pneumonia (caused by pneumococcal bacteria)	If you are 65 or older, you should get one dose of the PCV13, PCV15, or PCV 20 vaccine and at least one dose of the PPSV23 vaccine, depending on your age and health.
Shingles	If you are 50 or older, you need a zoster vaccination with the Shingrix vaccine, even if you have already had shingles or if you've had the older Zostavax vaccine. Shingrix requires two injections, given two to six months apart.
Tetanus, diphtheria, and pertussis	You need at least one tetanus shot, included with vaccines for diphtheria and pertussis in the combination Tdap vaccine, followed by a booster every 10 years.

Source: Centers for Disease Control and Prevention.

9. Learn to stress less

We all have stress in our lives. Whether it comes from a death in the family, the pain of arthritis, or a demanding job, the effects on the body are the same: the brain triggers a cascade of chemicals and hormones that speed heart rate, quicken breathing, increase blood pressure, and boost the amount of energy (in the form of blood sugar) supplied to muscles. Blood becomes “stickier” and more likely to clot, while the immune system releases inflammatory compounds. Intermittent episodes of stress are not harmful; in fact, you experience many of these symptoms during excitement and pleasure. However, chronic stress is a major contributor to a host of serious physical and psychological conditions.

Whatever sets your stress cycle in motion, having a personal escape plan can help. To create a

plan, make a list of the specific mental, emotional, and physical sensations that you feel when stressed. For example, do you get a stomachache, grind your teeth, raid the fridge, or have trouble thinking clearly? Whenever you notice your stress symptoms, take a moment—or two, or three—to relax. Here are some suggestions from the Benson-Henry Institute for Mind Body Medicine at Massachusetts General Hospital.

Learn relaxation skills. Deep breathing, meditation, progressive muscle relaxation, and visualization can evoke a state of rest and release. Here are two simple techniques:

- **Breath focus.** Place your hand just beneath your navel so you can feel the gentle rise and fall of your belly as you breathe. Breathe in slowly. Pause for a count of three. Exhale slowly. Pause for a count of three. Continue to breathe deeply for one

minute, pausing for a count of three after each inhalation and exhalation.

- **Body scan.** While sitting down, take a break from whatever you're doing and check your body for tension. Relax your facial muscles and allow your jaw to fall open slightly. Let your shoulders drop. Let your arms fall to your sides. Allow your hands to loosen so that there are spaces between your fingers. Uncross your legs or ankles. Feel your thighs sink into your chair, letting your legs fall comfortably apart. Feel your shins and calves become heavier and your feet grow roots into the floor. Now breathe in slowly and breathe out slowly.

Call a mental time out. If you tend to overreact to stressful situations, try this exercise the next time stress presents itself:

- **Stop.** By saying “Stop” or picturing a stop sign, you can halt the

negative reaction in its tracks.

- **Breathe.** Take a few deep breaths to reduce physical tension and step back from the problem before reacting.
- **Reflect.** Ask yourself how bad the situation really is. Is there another way to view it? What's the worst that could happen?
- **Choose.** Decide how to deal with the problem. Accept what you cannot change, and try to solve what you can.

Get moving. Regular aerobic exercise reduces levels of the body's stress hormones and stimulates the production of endorphins, chemicals in the brain that reduce pain and elevate mood. It also helps take your mind off whatever is causing your stress and puts you more “in the moment.” And it improves sleep, which is a stress-reducer in itself.

10. Skip most supplements

People used to think that you could compensate for dietary deficiencies by popping a multivitamin, but the benefits remain uncertain. The Women's Health Initiative concluded that postmenopausal women who took multivitamins did not have a lower death rate than non-users of vitamins and were just as likely to develop cardiovascular disease or cancers of the lung, colon and rectum, breast, and endometrium. Other studies have similarly found little or no evidence of protection against cardiovascu-



Practicing relaxation techniques, such as deep breathing, can help you keep your cool and make you less apt to overreact when stress strikes.



The best way to get the nutrients you need to stay healthy is by eating a balanced, healthy diet rather than relying on vitamins and other supplements.

lar disease or cancer from taking individual supplements such as vitamin E, vitamin C, beta carotene, or the popular B vitamin trio of B₆, B₁₂, and folic acid.

It is also very important to note that excessive use of supple-

the highest levels of calcium in the form of supplements may be more likely to have heart attacks or heart disease. Both observations, however, warrant further study.

Experts agree that the best way to get the nutrients you need

ments may be harmful. A Cochrane Collaboration review found that low-risk participants in trials for a host of diseases who were given supplements of vitamin A, vitamin E, and beta carotene had a slightly higher death rate. And evidence has been mounting in recent years that people who consume

is through food. No pill can make up for the fiber, antioxidants, and myriad other healthful compounds in fruits, vegetables, and legumes. It is likely that what counts is the synergistic interactions of these nutrients—which might also help explain why trials of single nutrients often don't pan out.

To be sure, there may be supplements worth considering. Calcium and vitamin D are essential for preserving bone density (see “Eat a balanced diet,” page 12). And if you are vegan or vegetarian, you may need to take vitamin B₁₂, an essential vitamin for the nerves, which is found in animal-based foods. Consult your doctor about what supplements may be right for you. ♥

Embracing the future

“You’re as young as you feel” is more than a platitude. A growing body of research has indicated that attitudes toward aging have a strong effect on health and longevity. People who have a positive view of aging are less likely to develop cardiovascular disease, dementia, and other chronic conditions. And a study of 6,500 seniors indicated that those who reported feeling at least three years younger than their calendar age had a significantly lower death rate over the next eight years compared with those who felt their age or older.

Researchers in the field of positive psychology have identified practices that can help you achieve and sustain a positive attitude toward your future. Here are a few quick suggestions:

Savor pleasure. Most people are primed to experience pleasure in special moments, such as a wedding day or a vacation. Everyday pleasures, on the other hand, can slip by without much notice. Slow down and focus on them. You will enjoy things more, whether it’s a tasty meal or a visit with a friend.

Practice gratitude. Try keeping a gratitude journal, in which you write down things for which you are grateful. Doing so will help you go through your days with greater appreciation. As you write, be specific and try to relive the sensations you felt as you remember what each thing means to you.

Cultivate mindfulness. Mindfulness involves opening fully to both pleasant and unpleasant experiences. By focusing on the here and now, many people who practice mindfulness find that they are less likely to get caught up in worries about the future or regrets over the past and are better able to form deep connections with others.



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For some, adopting a more positive attitude comes naturally with age. If it doesn’t, keeping a gratitude journal and practicing mindfulness may help shift your outlook so you become more resilient as well as content.

Retain a sense of purpose. It’s normal to experience a sense of loss and even grief when your children leave home, you retire, or you lose a loved one. Aim to view such a life change as a new challenge. Focus on what’s possible. Perhaps you can organize a family reunion or plunge into a volunteer activity you haven’t previously had time for.

Emphasize the positive. It’s a key factor in acquiring resilience—the ability to bounce back from bad times. You may find that adopting a positive attitude comes naturally with age. Studies have shown that people over 60 tend to brood less, perhaps because the amygdala (the area of the brain that consolidates emotion and memory) is less responsive to negatively charged situations. On survey after survey, older people report that they have become more satisfied with their lives as they have aged, despite the losses that accumulate with passing years. This is most likely because they tend to minimize the negative, accept their limitations, and set reasonable goals for the future. ♥

Resources

Organizations

Bone Health and Osteoporosis Foundation

251 18th St. S, Suite 630
Arlington, VA 22202
800-231-4222 (toll-free)
info@bonehealthandosteoporosis.org
www.bonehealthandosteoporosis.org

A nonprofit organization that supports research on osteoporosis and develops educational programs and materials.

North American Menopause Society

30050 Chagrin Blvd, Suite 120W
Pepper Pike, OH 44124
440-442-7550
info@menopause.org
www.menopause.org

A nonprofit organization devoted to promoting the understanding of menopause and improving the health of women at perimenopause and beyond.

Office on Women's Health

Department of Health and Human Services
1101 Wootton Pkwy
Rockville, MD 20852
800-994-9662 (toll-free)
www.womenshealth.gov

The federal government's clearinghouse for women's health information. The website is a gateway to a trove of reliable, free information on scores of topics.

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(Harvard Medical School, 2022)

Alzheimer's Disease: A guide to diagnosis, treatment, and caregiving
(Harvard Medical School, 2021)

Better Balance: Simple exercises to improve stability and prevent falls
(Harvard Medical School, 2022)

Better Bladder and Bowel Control: Practical strategies for managing incontinence
(Harvard Medical School, 2021)

Boosting Your Energy: How to jump-start your natural energy and fight fatigue
(Harvard Medical School, 2020)

Controlling Your Blood Pressure: What to do when your doctor says you have hypertension
(Harvard Medical School, 2021)

Coping with Hearing Loss: A guide to prevention and treatment
(Harvard Medical School, 2023)

Exercises for Bone Strength: 7 workouts to help prevent osteoporosis and keep you standing tall
(Harvard Medical School, 2020)

The Harvard Medical School 6-Week Plan for Healthy Eating
(Harvard Medical School, 2023)

Healthy Feet: Preventing and treating common foot problems
(Harvard Medical School, 2022)

Improving Memory: Understanding age-related memory loss
(Harvard Medical School, 2022)

Improving Sleep: A guide to a good night's rest
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(Harvard Medical School, 2024)

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(Harvard Medical School, 2023)

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Positive Psychology: Harnessing the power of happiness, mindfulness, and inner strength
(Harvard Medical School, 2023)

Preventing Falls
(Harvard Medical School, 2021)

Sexuality in Midlife and Beyond: Reclaiming your sexual vitality and putting the fun back into sex
(Harvard Medical School, 2023)

Skin Care and Repair: Options for preventing and treating skin damage and common skin conditions
(Harvard Medical School, 2024)

Thyroid Disease: Understanding hypothyroidism and hyperthyroidism
(Harvard Medical School, 2022)



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Back Pain	Headaches	Sensitive Gut
Balance	Hearing Loss	Sexuality
Body-Weight Exercise	Heart Disease	Shoulder Pain Relief
Cardio Exercise	High Blood Pressure	Skin Care
Caregiving	Incontinence	Sleep
Cholesterol	Inflammation	Starting to Exercise
Cognitive Fitness	Interval Training	Strength Training
Combating Loneliness	Joint Pain Relief	Stress Management
COPD	Knees & Hips	Stretching
Core Workout	Leg Pain	Stroke
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